Lab 01 Manual

Haleemah Anwary

Seneca College

SRT411NAA: Digital Data Analytics

Kosalan Balarajah

Date Submitted: May 21st, 2024.

Table of Contents

Pivot Table Tasks (Task 01) 4

1) Create a pivot table using the “Lab_01_Sales” provided. 4

2) Extract countries using pivot tables. 4

3) Create a rollup of States and Countries using pivot tables. 5

4) Create a view that shows countries and their respective revenue. 5

5) Expand the view to show the countries, states, and the revenue. 6

6) Using Value Field Settings in Pivot Value create a Sum of Revenue Per Country. Show the Value as: 6

7) Create a full Revenue, Cost, Sum of Profit Cost Analysis using Pivot Table 7

8) Create a Calculated Field Column using “PivotTable Analyze” Wizard “Fields, Items and Sets” Function. Name the column “Calculated Profit”. 8

9) Are the Calculated Profit and Sum of Profit matching? 8

10) Create a filter for Gender and show the Cost Analysis for “M” and “W”. 8

11) Using the Pivot Table mechanism you have learned, derive a time-based summary table that explains how the countries are doing over the periods of time analysed. 9

Formulas Tasks (Task 02) 10

MAX 10

MIN 11

IF 12

IFS 14

LEN 14

LEFT 15

RIGHT 15

TEXT (Date to Text) 16

TRIM 17

SUBSTITUTE 17

SUM 18

SUMIF 19

SUMIFS 20

COUNT 21

COUNTIF 22

DAYS 22

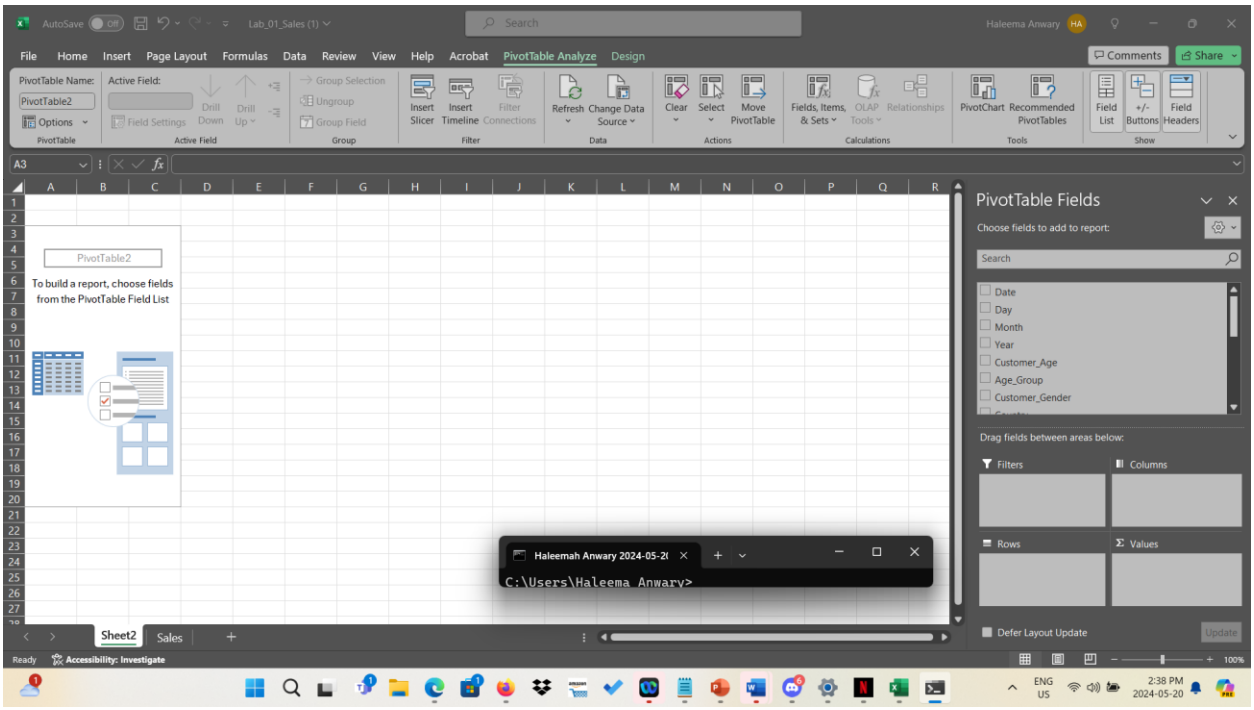
NETWORKDAYS 23

XLOOKUP Tasks (Task 03) 24
XLOOKUP 24
XLOOKUP Multiple Rows 25
XLOOKUP Exact Match 25
XLOOKUP Search Order 26
XLOOKUP Horizontal 26
XLOOKUP With Sum 27
VLOOKUP 27
Conditional Formatting (Task 04) 28
Example 1: Greater than 28
Example 2: Data bars 29
Example 3: Color scales 29
Example 4: Between using Date column 30

Pivot Table Tasks (Task 01)

1) Create a pivot table using the “Lab_01_Sales” provided.

Go to Insert tab > click on PivoTable> Select the range of your data > choose where you want the report to be placed.



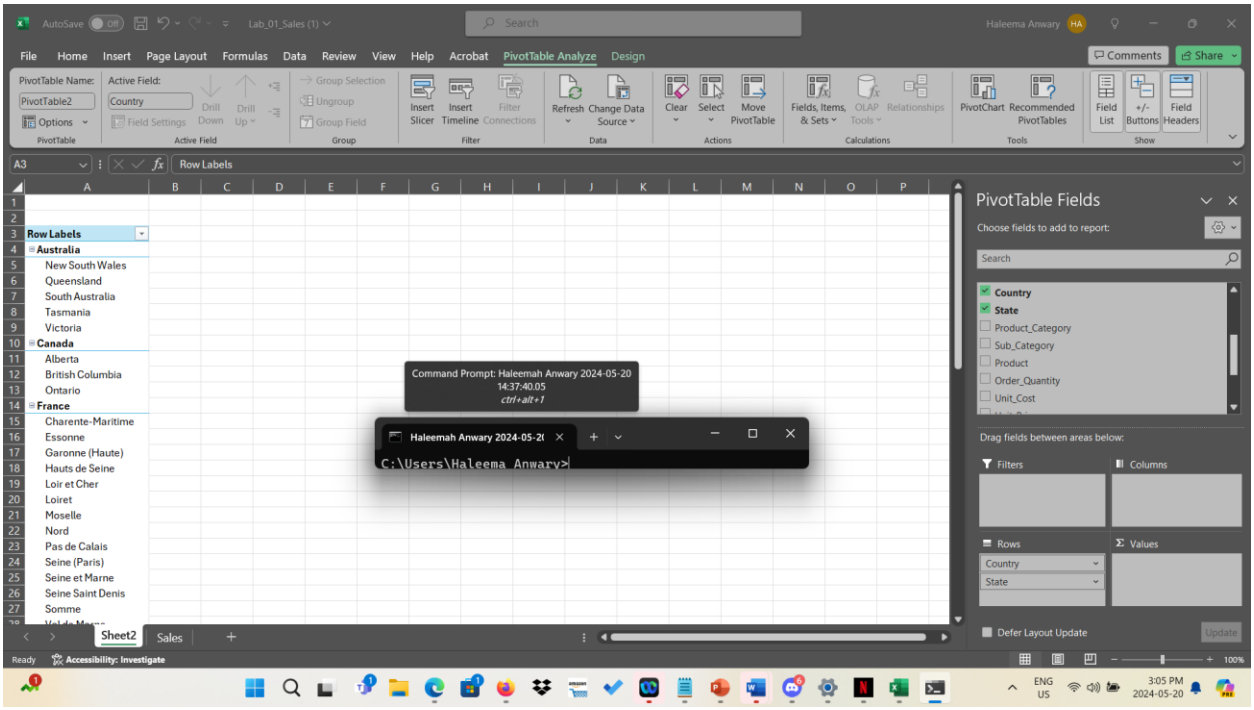
2) Extract countries using pivot tables.

Drag the country filed to the Rows area.



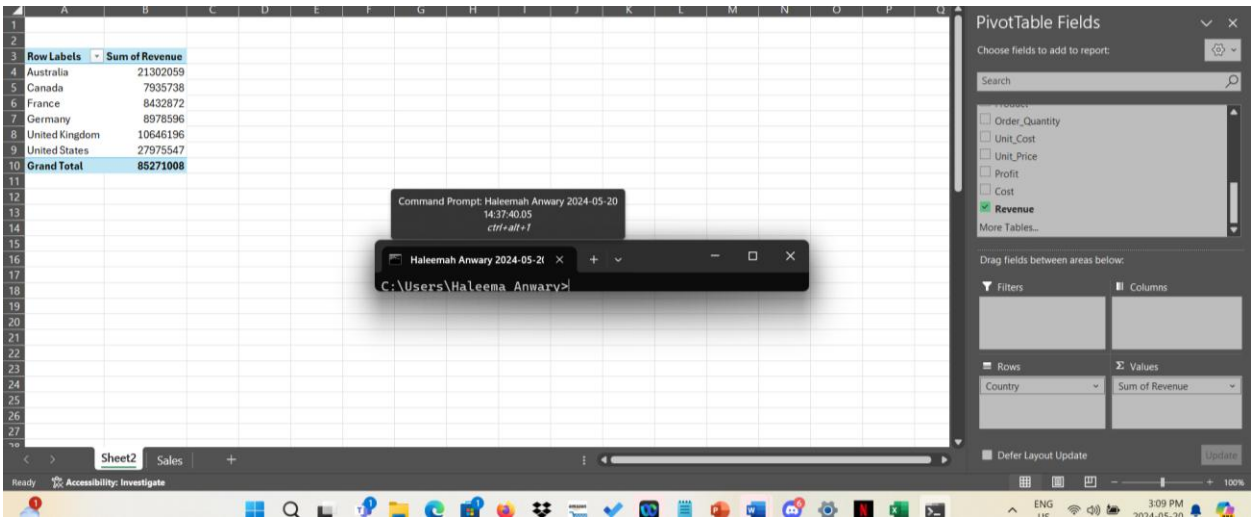
3) Create a rollup of States and Countries using pivot tables.

Drag the State field to the Rows under the Country to get a rollup of States and Countries using pivot tables.



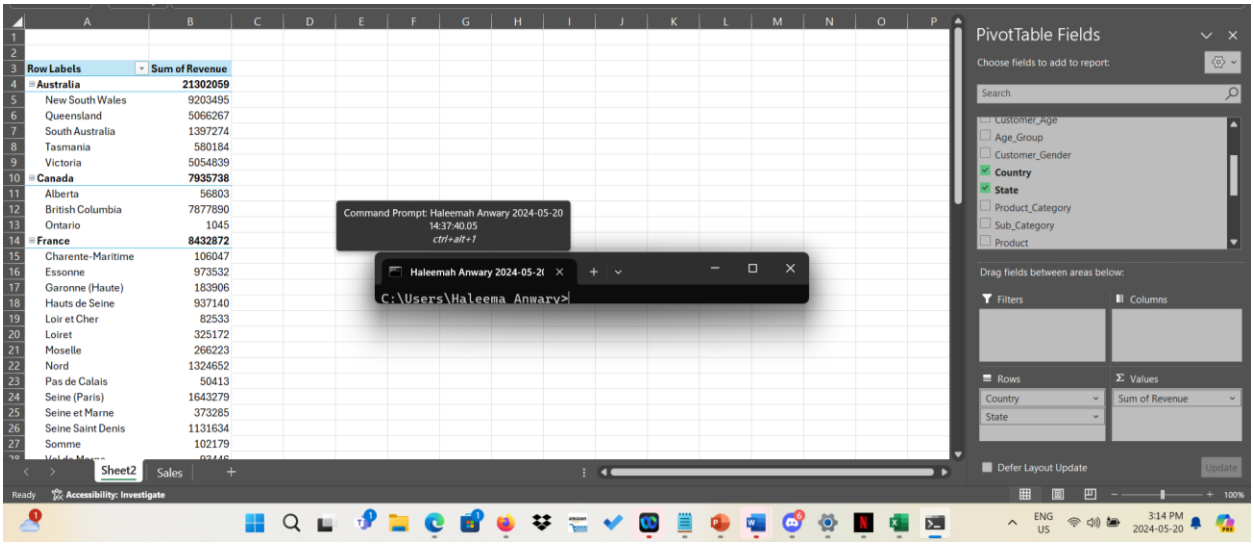
4) Create a view that shows countries and their respective revenue.

Drag the country field to the Rows area and Drag the Revenue field to the values area to get the total revenue for each country.



5) Expand the view to show the countries, states, and the revenue.

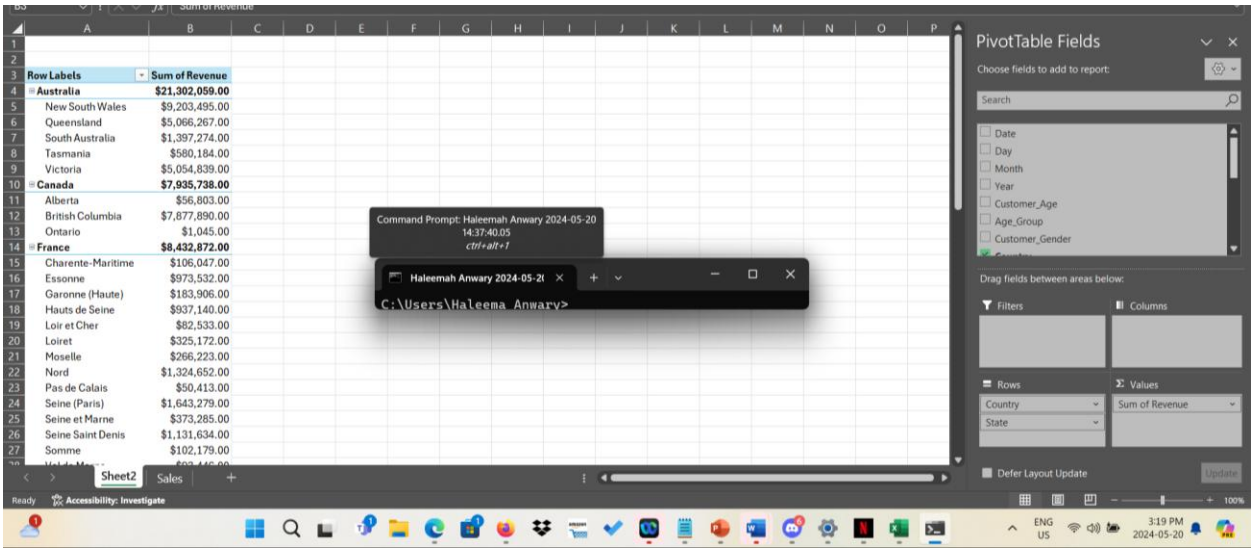
Drag the State field to the rows area under the Country again and this will nest the states under their respective countries and show the revenue.



6) Using Value Field Settings in Pivot Value create a Sum of Revenue Per Country. Show the Value as:

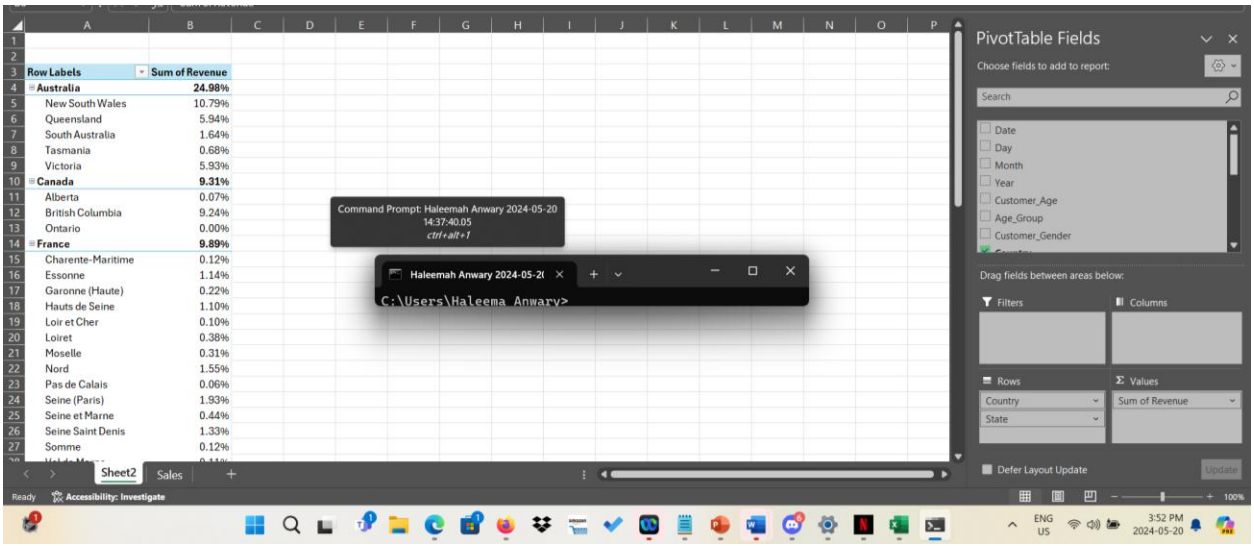
a) As Currency

Here we can click on sum of Revenue in the pivot table, go to the value Field settings, click on Number Format, and choose currency and click ok to show the total value as currency.



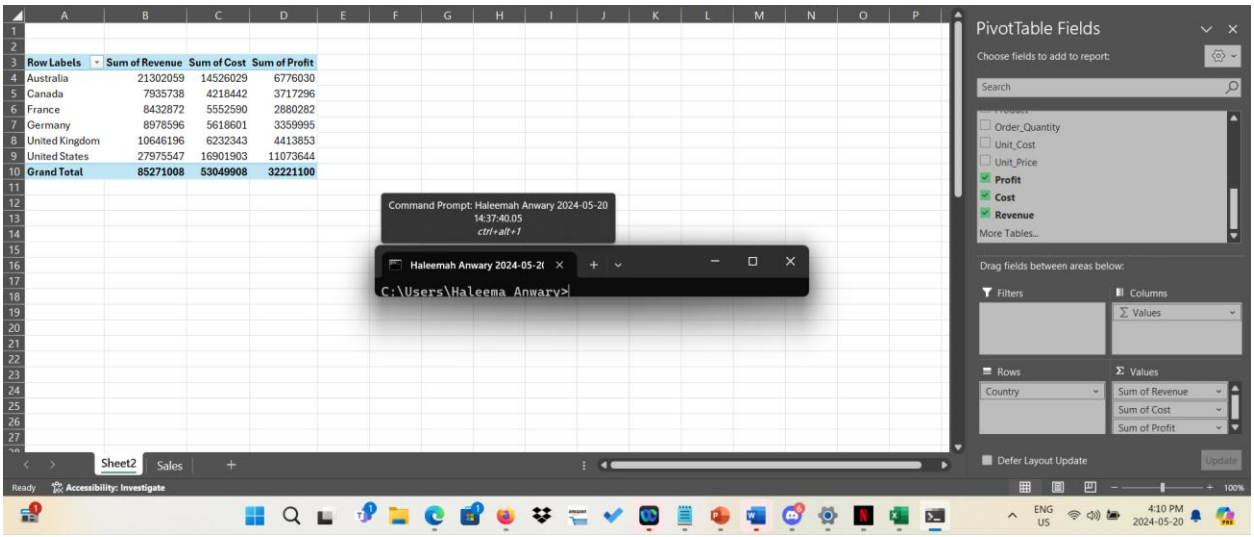
b) As Percentage

Here we click on the sum of the Revenue in the pivot table and go to the Value Field settings and click on show values As and select % of Grand Total.



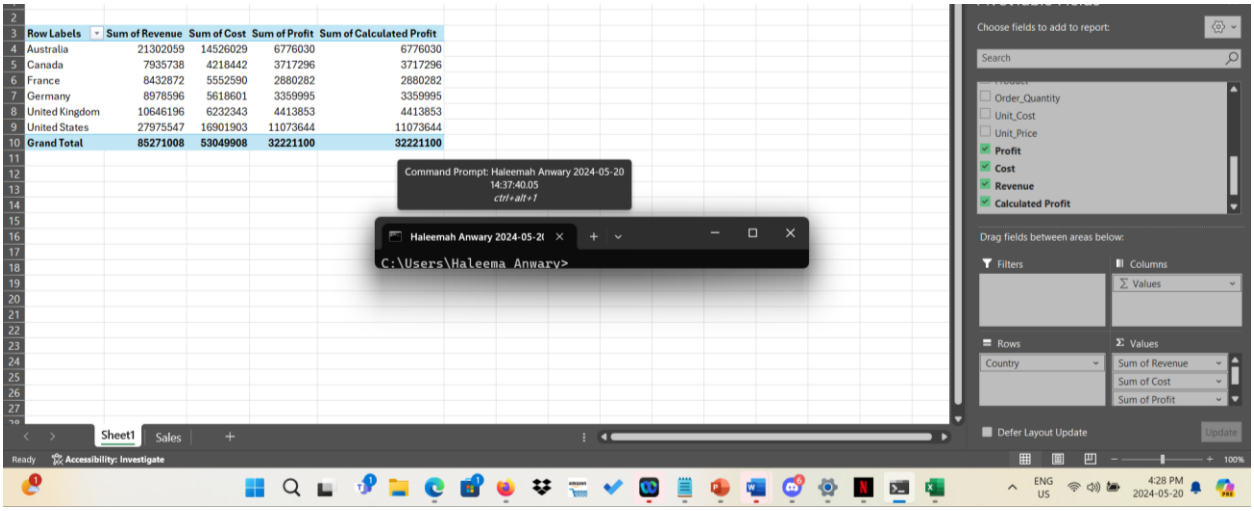
7) Create a full Revenue, Cost, Sum of Profit Cost Analysis using Pivot Table

Here we drag the country field to the Rows area and the Revenue, cost, and profit to the values area.



8) Create a Calculated Field Column using “PivotTable Analyze” Wizard “Fields, Items and Sets” Function. Name the column “Calculated Profit”.

Here we must go “PivotTable Analyze tab” and click on the “Fields, Items, and Sets” and then select “Calculated Field”. In the dialog box, enter the name as “Calculated Profit” and enter “=Revenue-cost” in the formula field, click add and ok to create a Calculated Field Column.

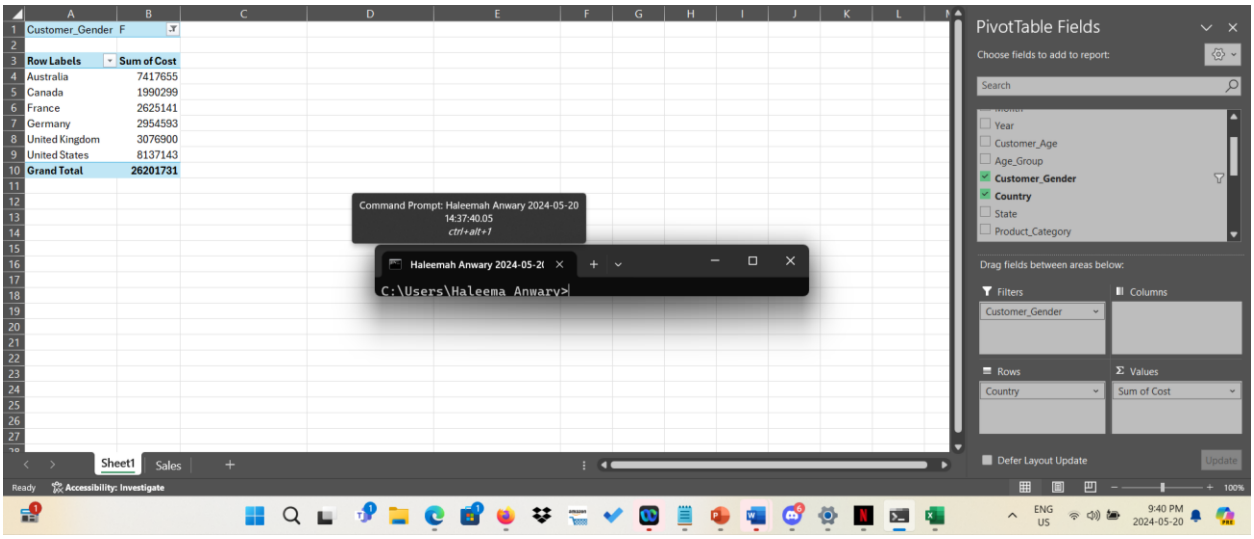


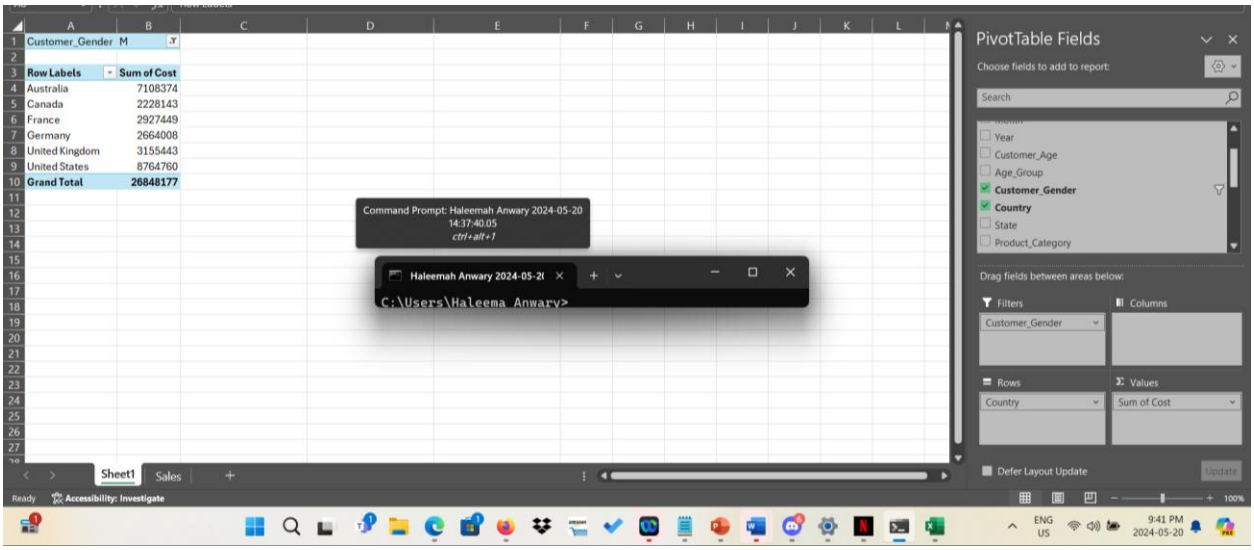
9) Are the Calculated Profit and Sum of Profit matching?

Yes, the screenshot above proves that the Calculated Profit matches the Sum of Profit.

10) Create a filter for Gender and show the Cost Analysis for “M” and “W”.

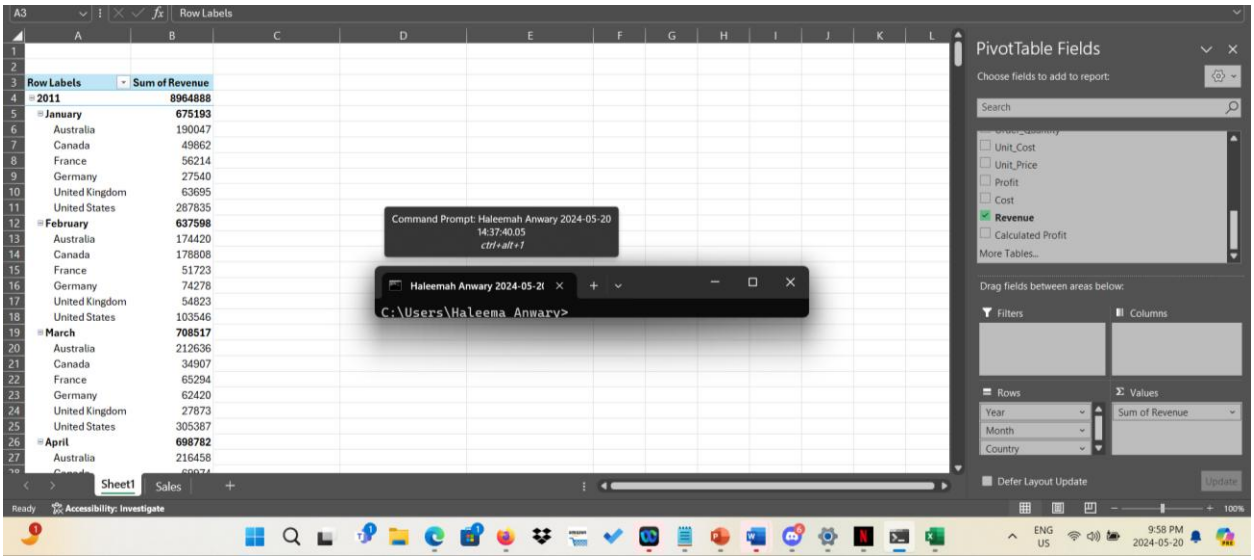
Here we need to drag the Customer_Gender field to the filters area and drag the cost field to the Values area. We can use the filter dropdown to select M or W and that’s how we can see the cost analysis for each gender.





11) Using the Pivot Table mechanism you have learned, derive a time-based summary table that explains how the countries are doing over the periods of time analysed.

To do this, we can drag the year and Month fields to the Rows area, drag the country field to the rows area under Year and Month and finally drag the Revenue field to the Values area. Following this, we can see the revenue for each country over the years and months.



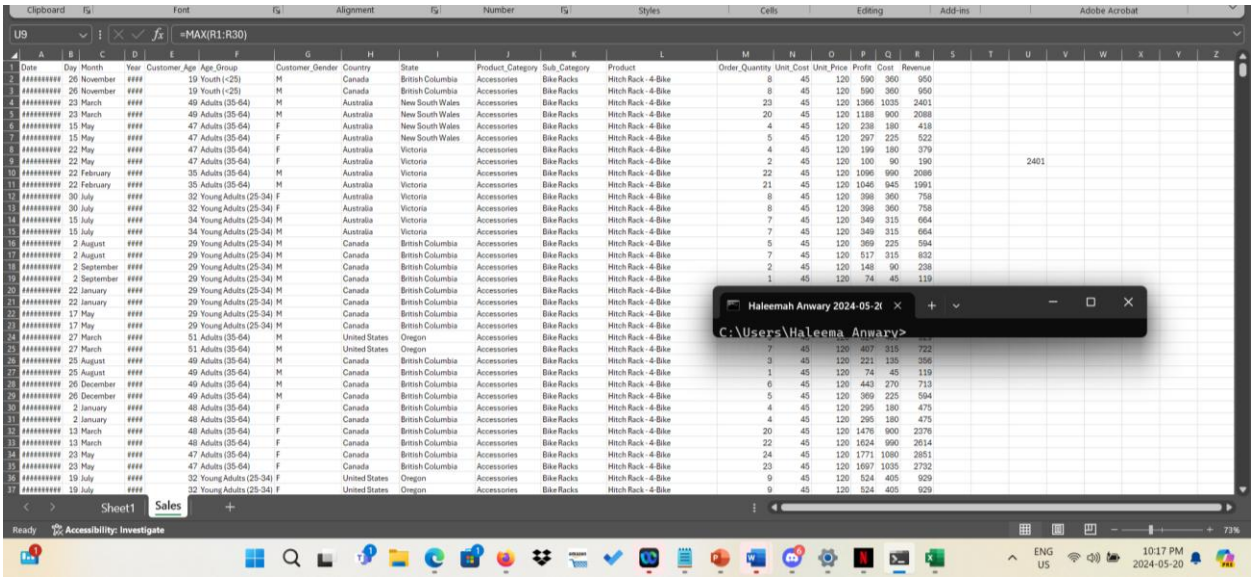
Formulas Tasks (Task 02)

MAX

The 'MAX' function returns the largest number in a set of values.

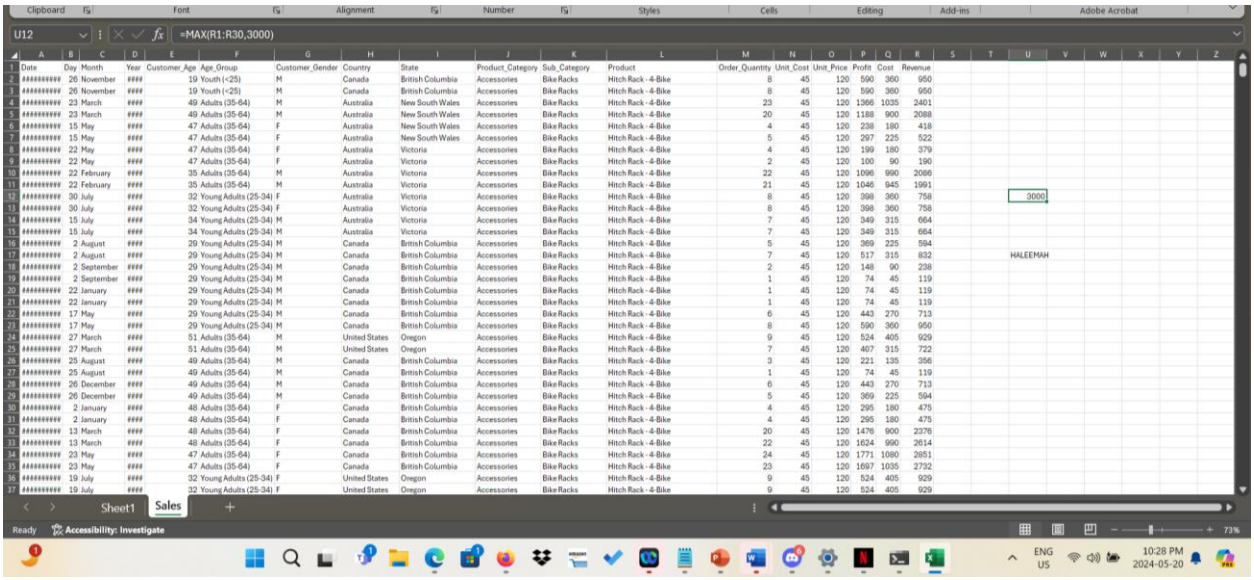
Example 1:

The formula =MAX (R1:R30) gives us the largest value in the range R1:R3



Example 2:

The formula =MAX(R1:R30,3000) gives us largest value in the range R1:R30 and the value 3000.

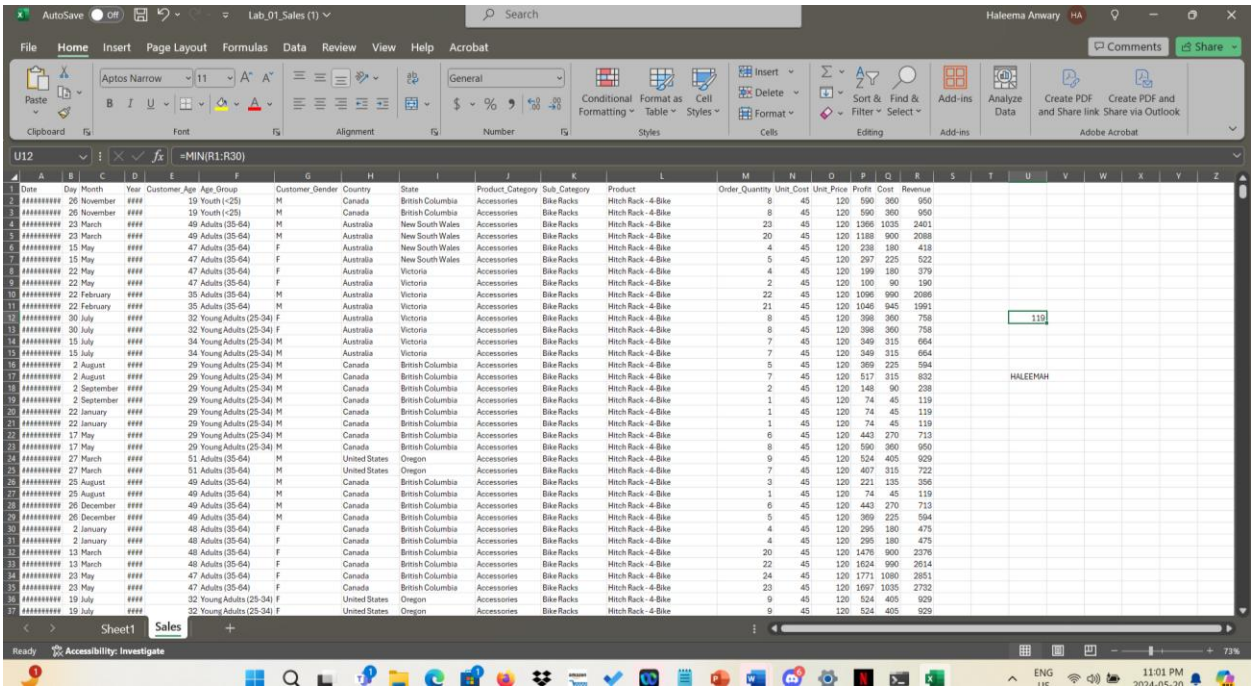


MIN

The ‘MIN’ function returns the smallest number in a set of values.

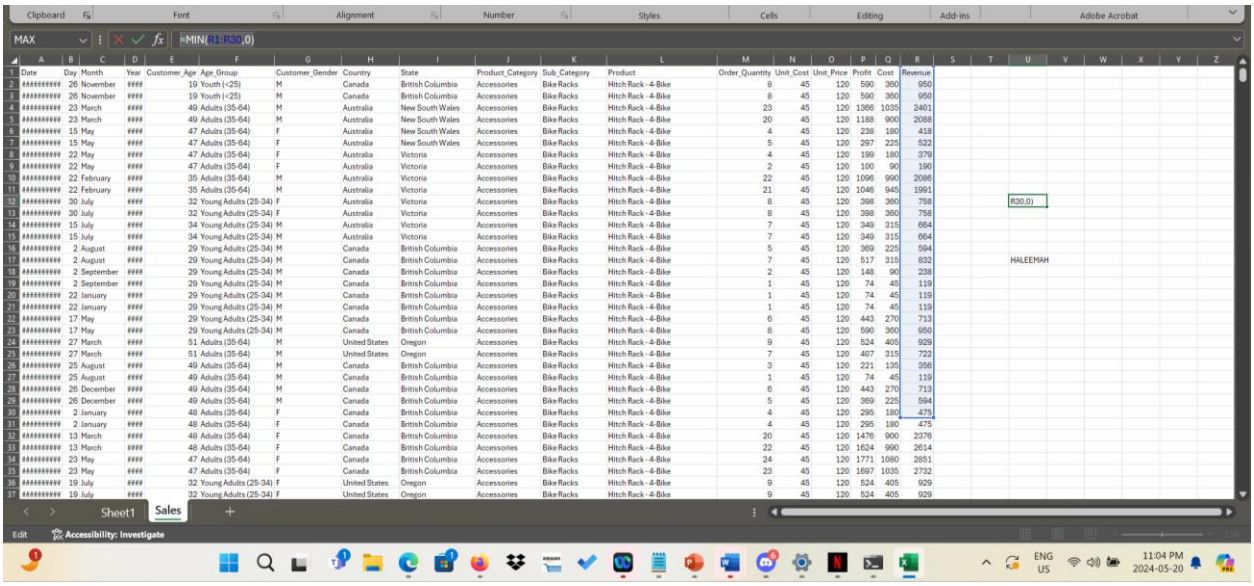
Example 1:

In this example the formula =MIN(R1:R30) return the smallest numbers in the range R1:R30.



Example 2:

In this example the formula =MIN(R1:R30,0) returns the smallest numbers in the range R1:R30 and 0.

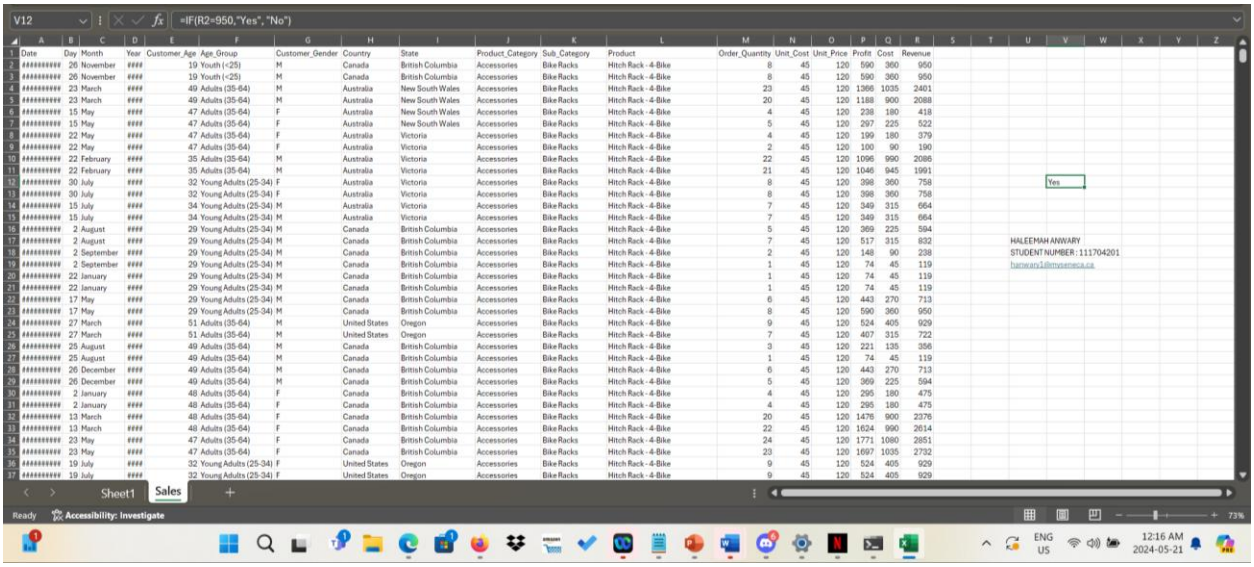


IF

The IF function checks whether a condition is met and returns one value if true and another value if false.

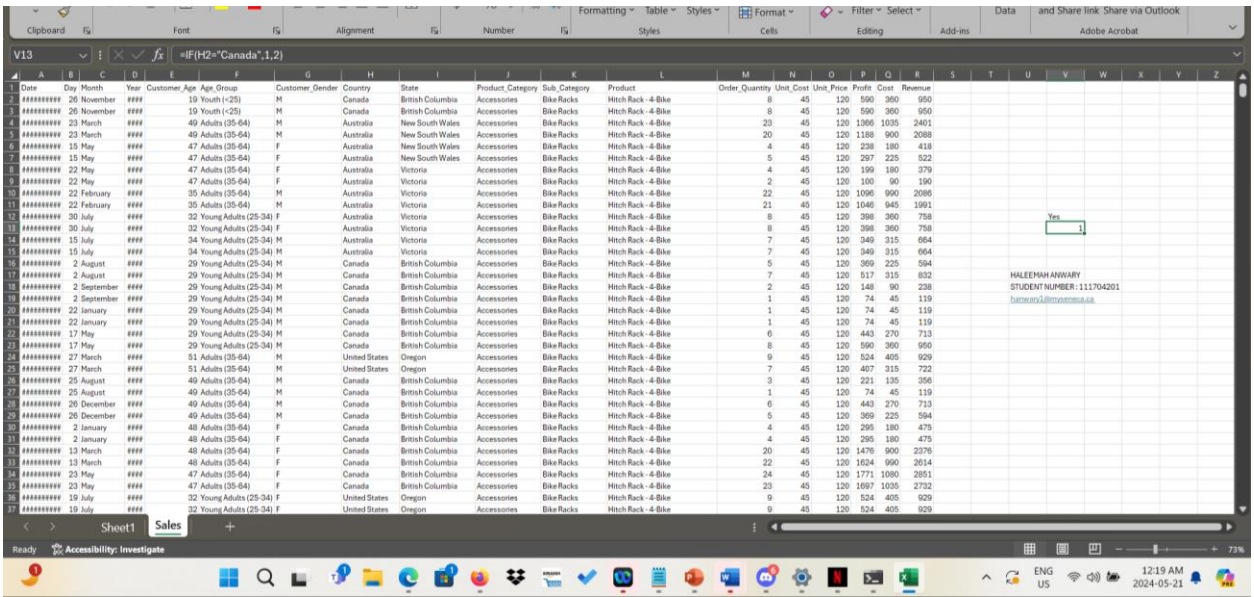
Example 1: =IF(R2=950,"Yes","No")

The formula in cell V12 says: *IF(R2 = 950, then return Yes, otherwise return No)*.



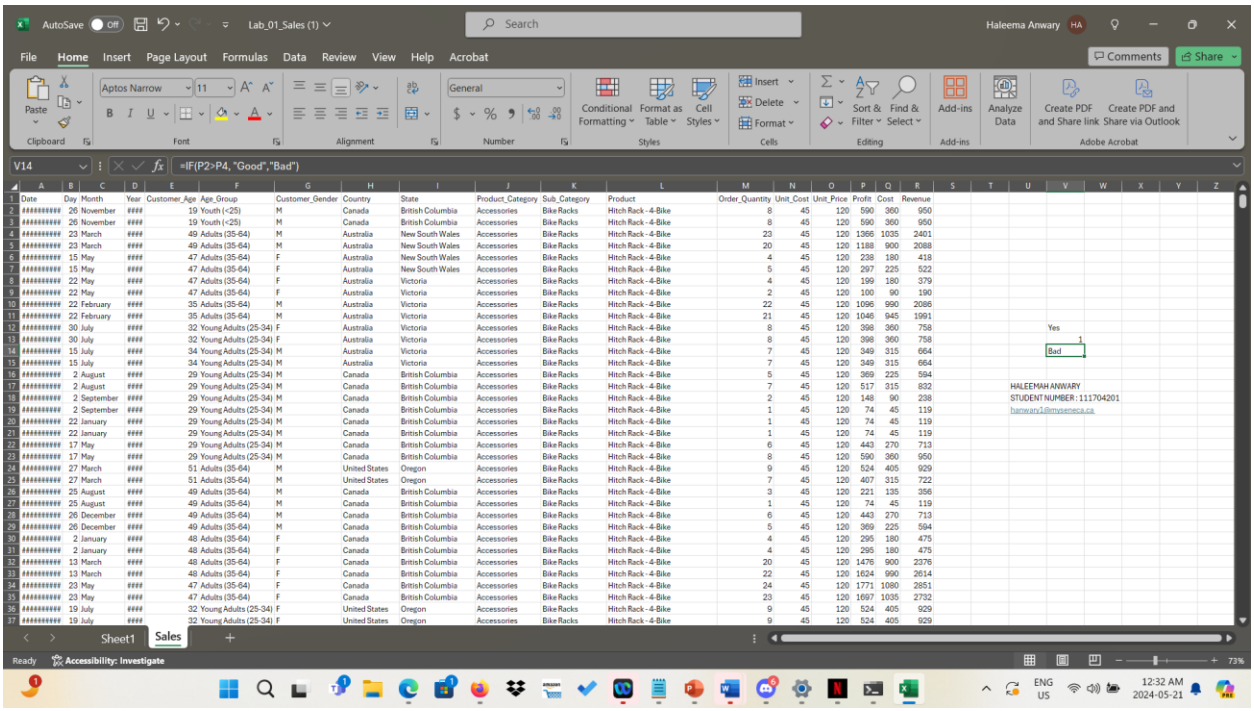
Example 2: =IF(H2="Canada",1,2)

It is used to evaluate both text and values. In this example, cell V13 says: *IF(H2 = Canada, then return a 1, otherwise return a 2)*.



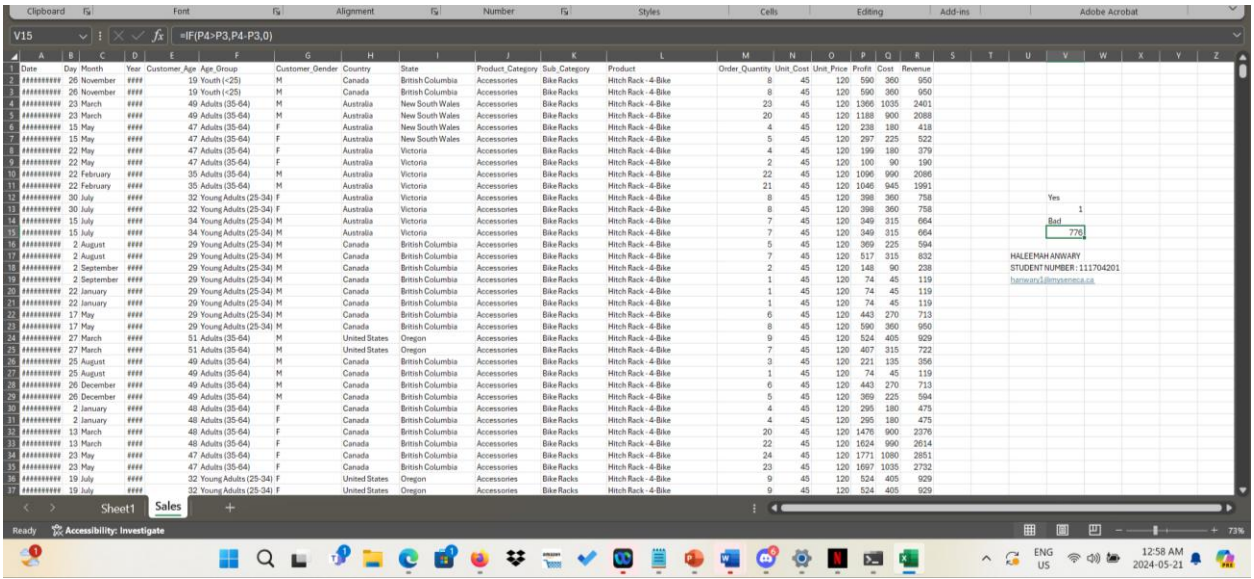
Example 3: =IF(P2>P4, "Good","Bad")

It can also be used for mathematical operators. In this example, the IF function in V14 is saying *IF (P2 Is Greater Than P4, then return "Good", otherwise return "Bad").*



Example 4: =IF(P4>P3,P4-P3,0)

It can also be used to do mathematical calculations. So the formula in V15 is saying *IF(P4 is Greater than P3, then Subtract the P3 amount from the P4 amount, otherwise return nothing).*

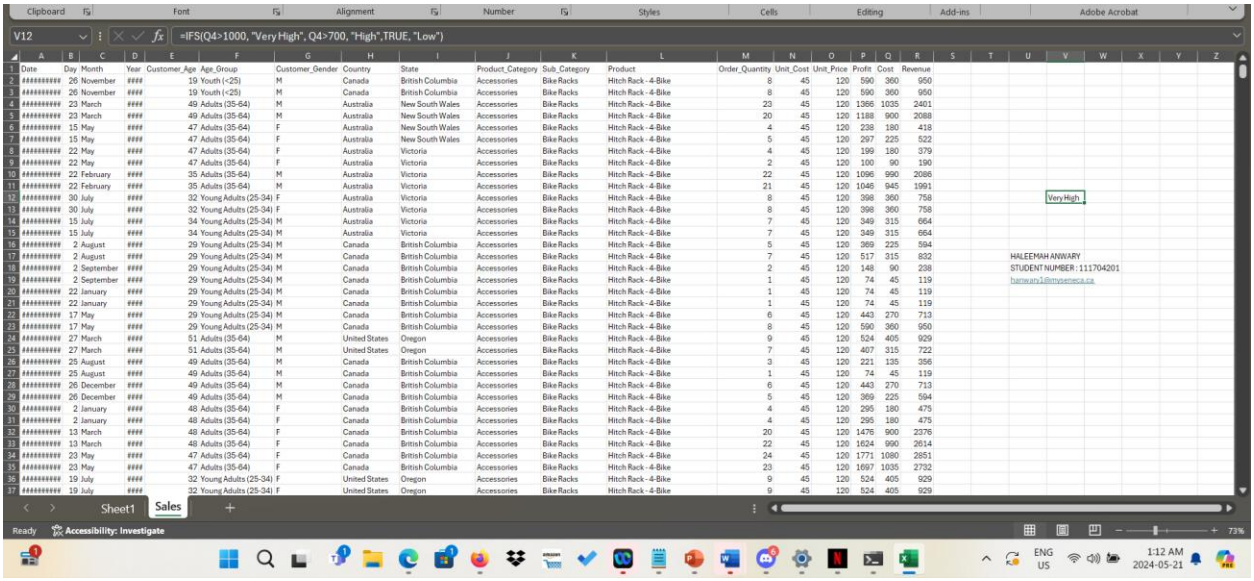


IFS

The ‘IFS’ function checks whether one or more conditions are met and returns a value corresponding to the first TRUE condition.

Example 1: =IFS(Q4>1000, "Very High", Q4>700, "High",TRUE, "Low")

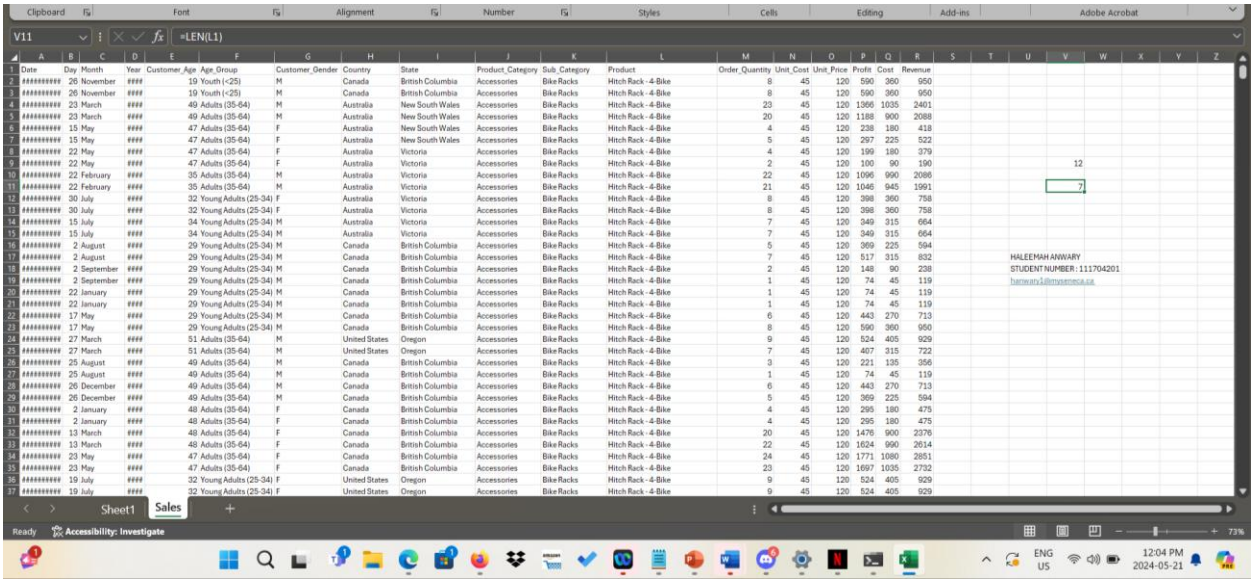
This formula says IF (Q4 is Greater Than 1000, then return "Very High", IF (Q4 is Greater Than 700, then return "High", For all other values (Q4 is 700 or less), return "Low."



LEN

The ‘LEN’ function returns the number of characters in a text string.

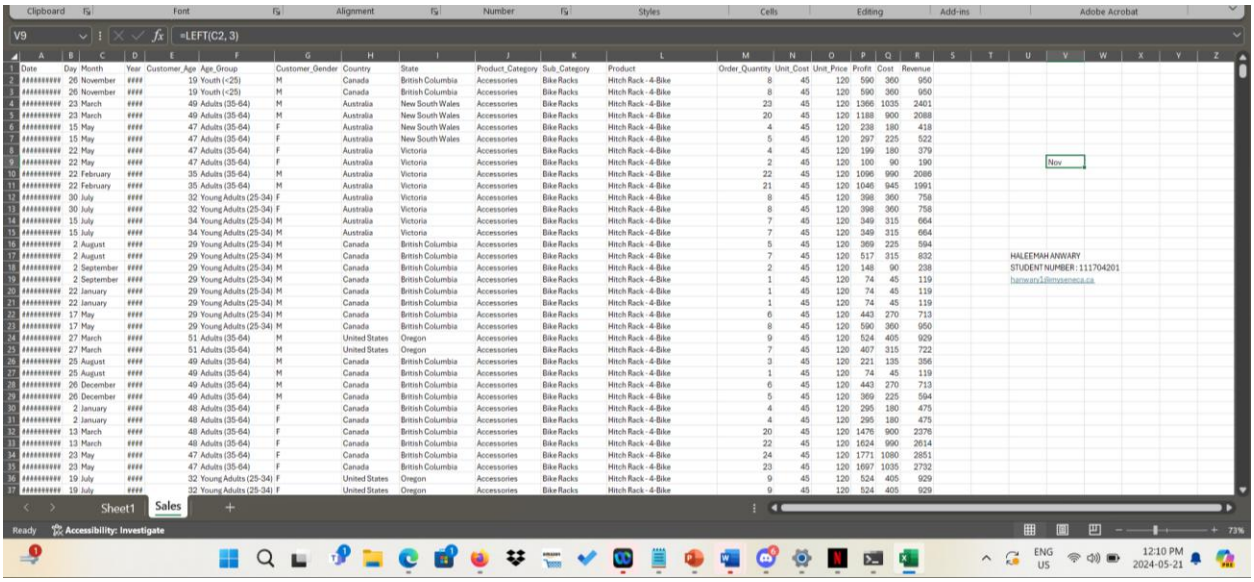
In this example I used the formula =LEN(L1) which is “Product” which has 7 characters.



LEFT

The ‘LEFT’ function returns the first character or characters in a text string, based on the number of characters you specify.

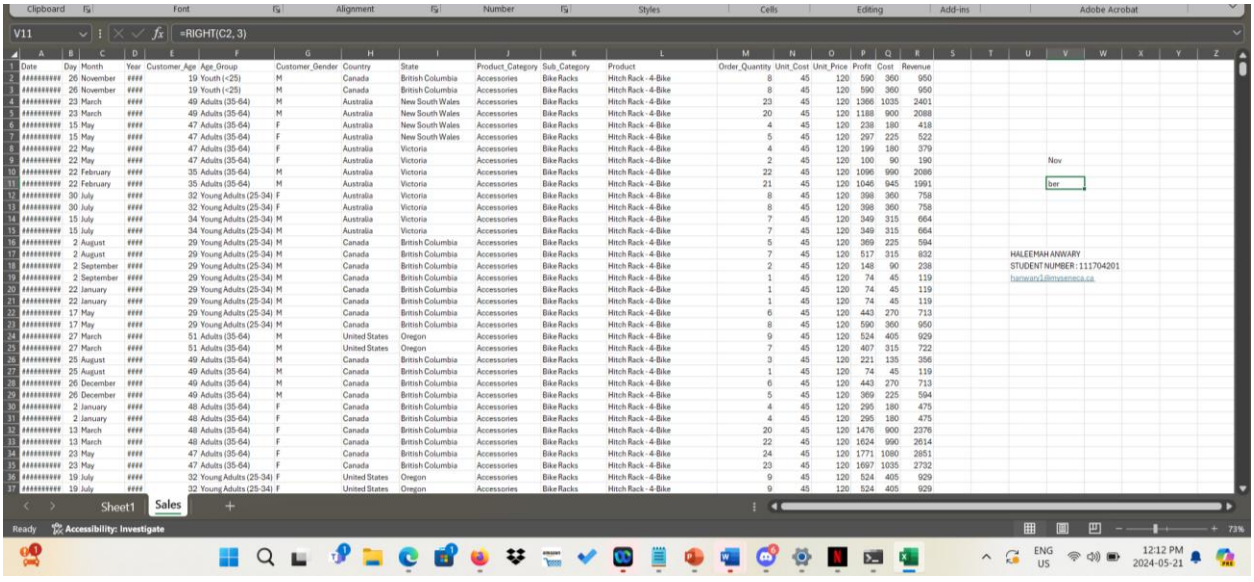
In this example I used the formula =LEFT(C2, 3). C2 spells ‘November’ and I want to get only the first 3 characters.



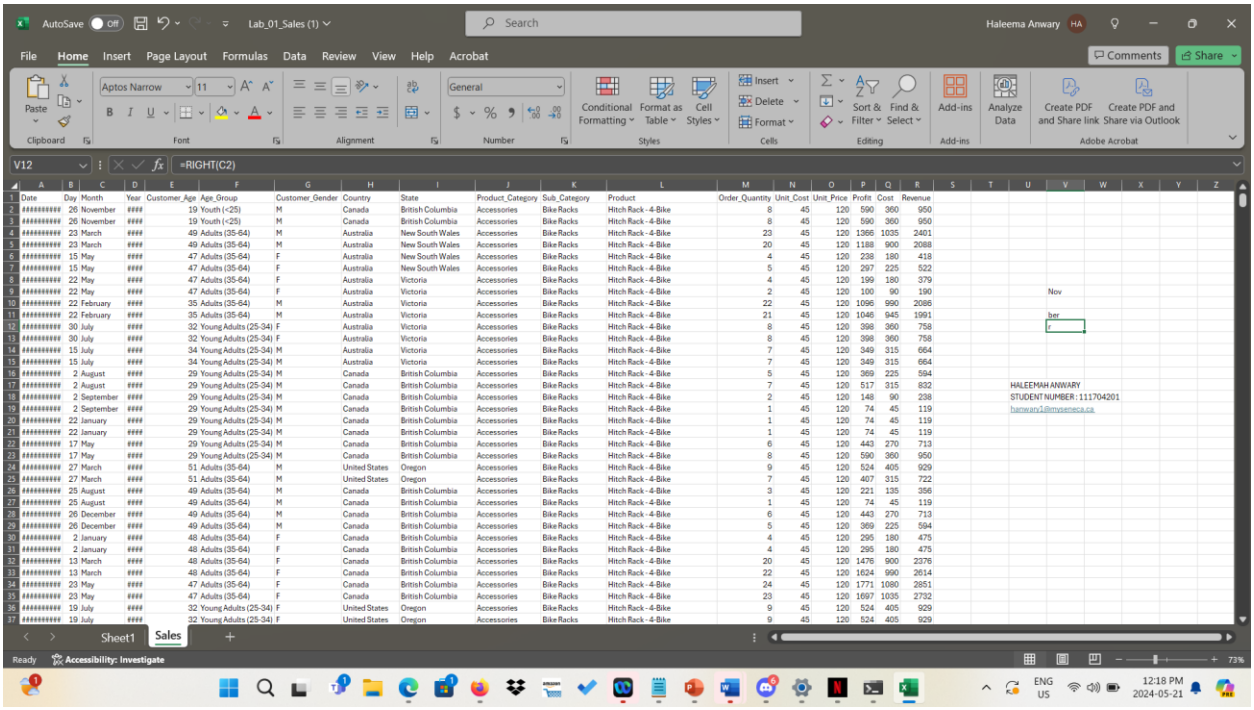
RIGHT

The ‘RIGHT’ function returns the last character or characters in a text string, based on the number of characters you specify.

In this example I used the formula =RIGHT(C2, 3). As a result of this formula, we get ‘ber’ as that’s the last 3 characters.



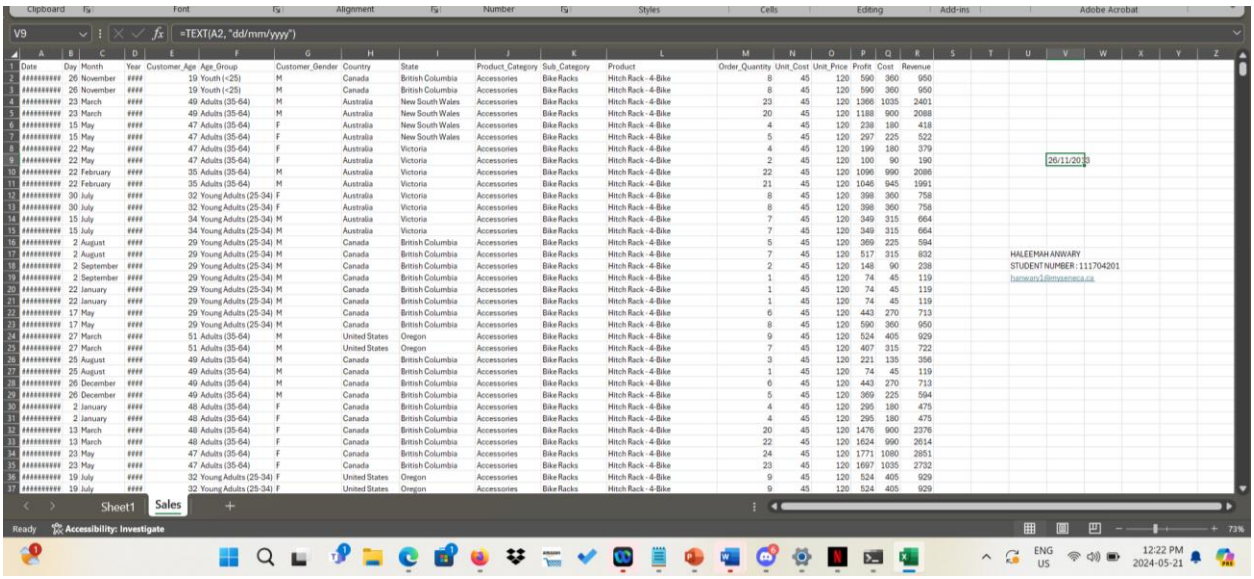
Example 2: In this example I used the formula =RIGHT(C2). As a result we will get just the last character of the string ‘November’ which is ‘r’.



TEXT (Date to Text)

The ‘TEXT’ function converts a value to text in a specific number format.

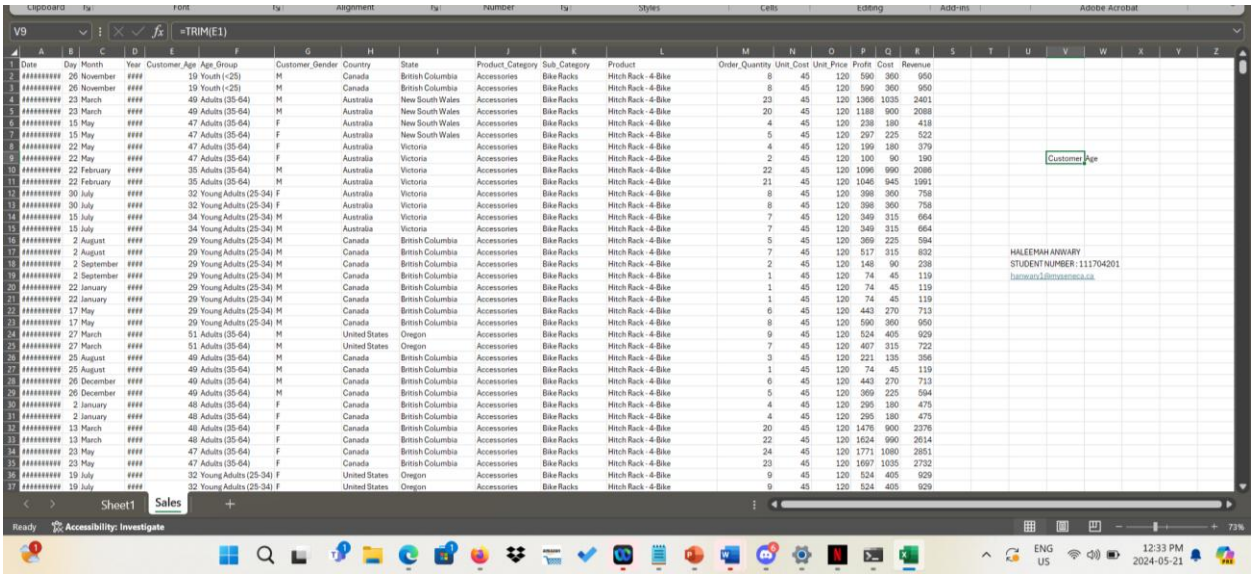
In the example below I used the formula =TEXT(A2, "dd/mm/yyyy").



TRIM

The ‘TRIM’ function removes all spaces from text except for single spaces between words.

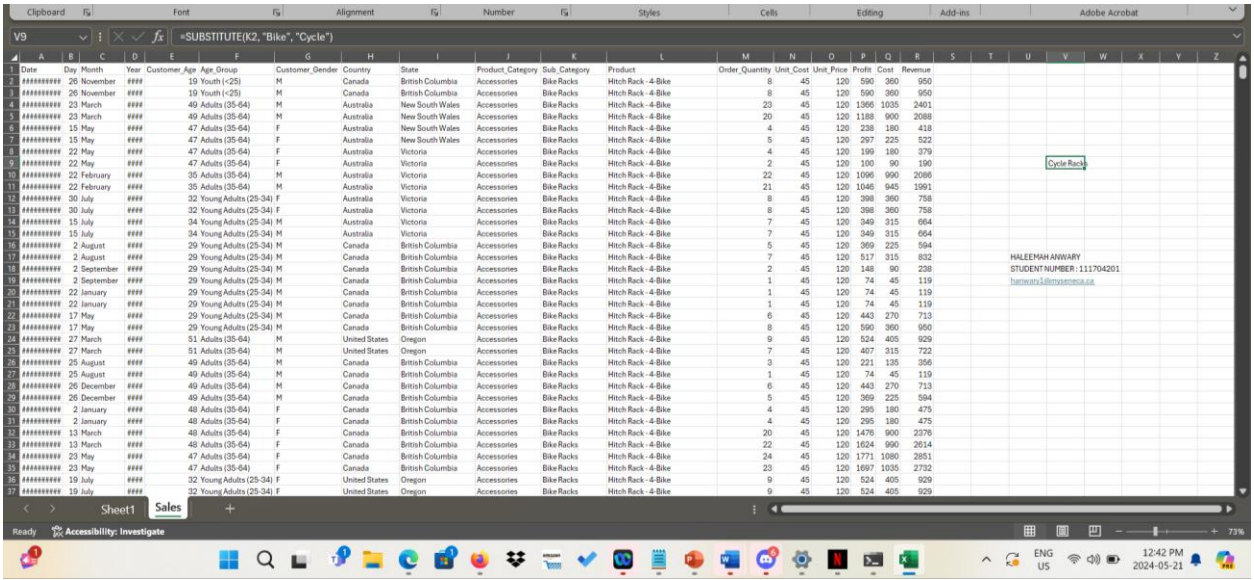
In this example, the formula used is shown but nothing has changed because there’s no irregular spacing.



SUBSTITUTE

The ‘SUBSTITUTE’ function substitutes new text for old text in a text string.

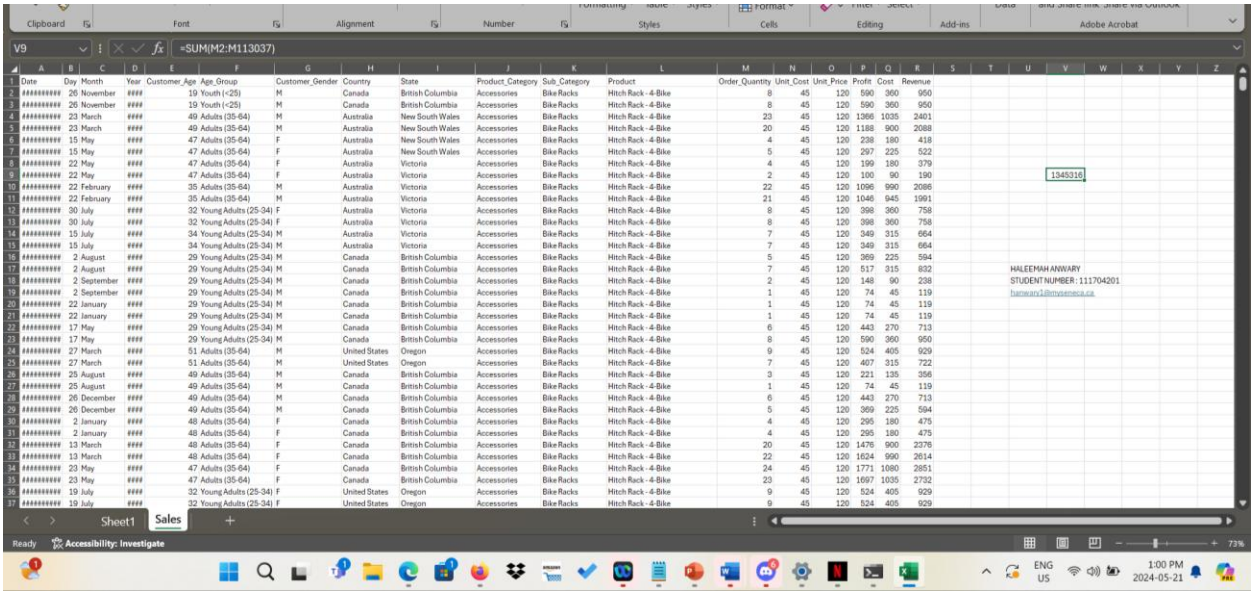
In this example I used the formula =SUBSTITUTE(K2, "Bike", "Cycle") . K2 is originally ‘Bike Racks’ but we want to substitute Bike to Cycle so we can use this formula to do so. The result will be “Cycle Racks.”



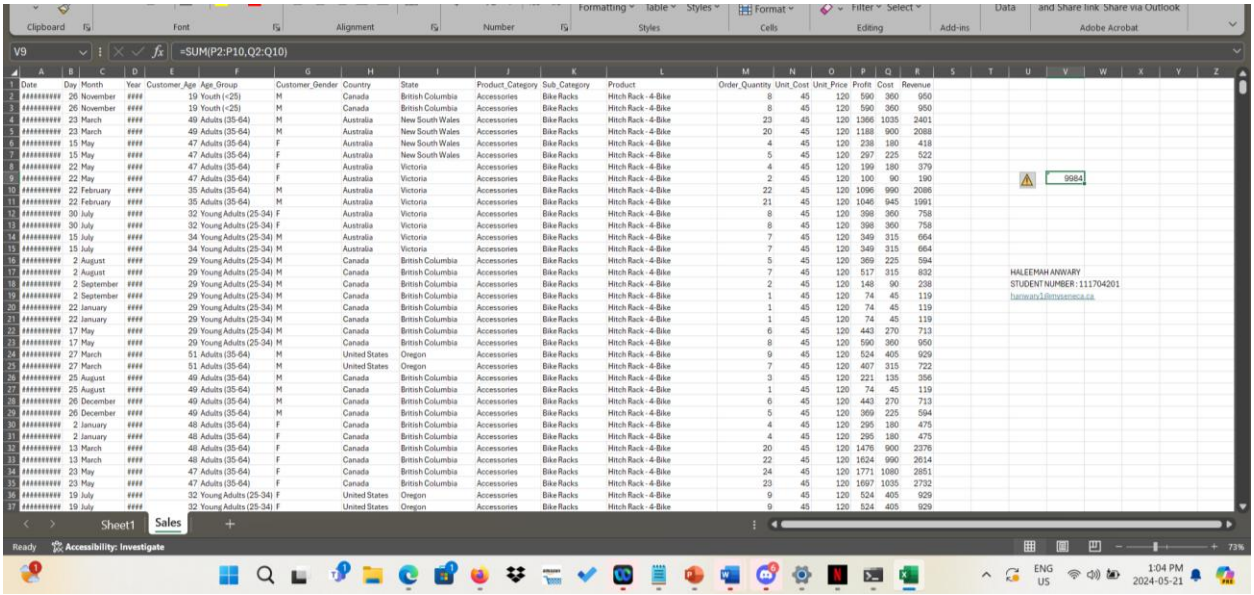
SUM

The ‘SUM’ function adds all the numbers in a range of cells.

Example 1: =SUM(M2:M113037) this formula adds the values in cells M2:M113037.



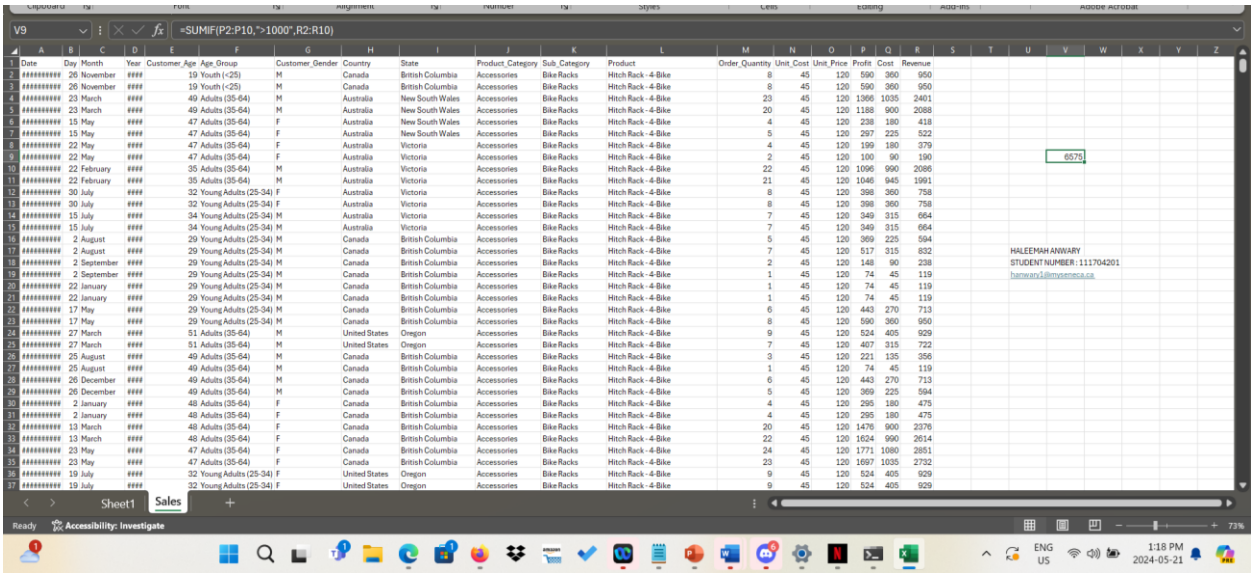
Example 2: =SUM(P2:P10,Q2:Q10) this formula adds the values in cells P2:P10, as well as cells Q2:Q10.



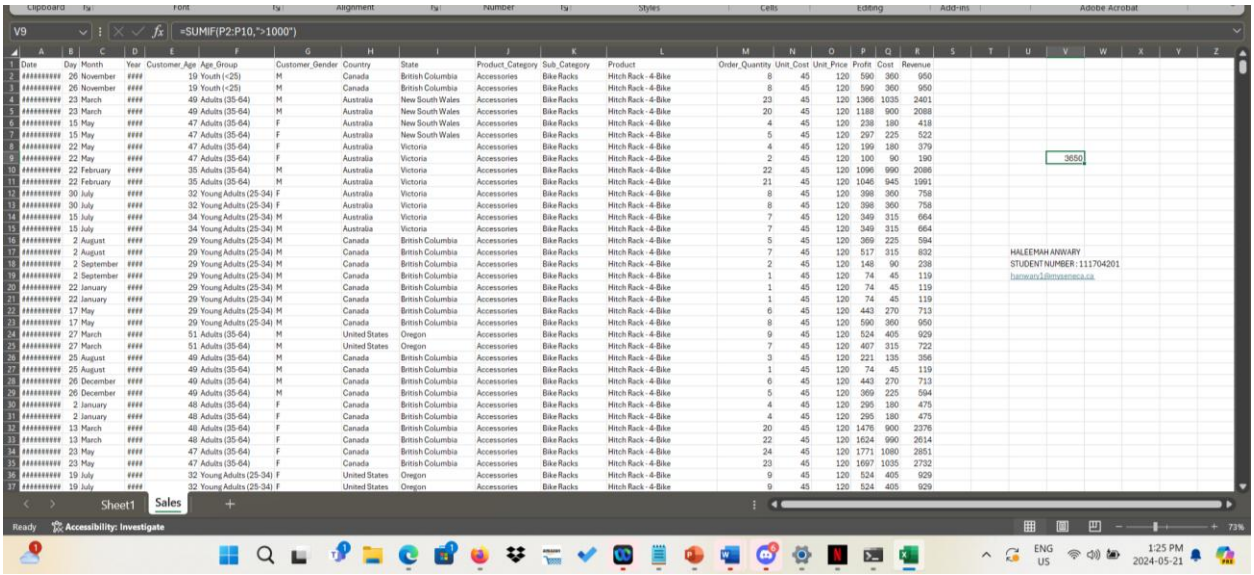
SUMIF

The ‘SUMIF’ function adds all the numbers in a range of cells based on a single condition.

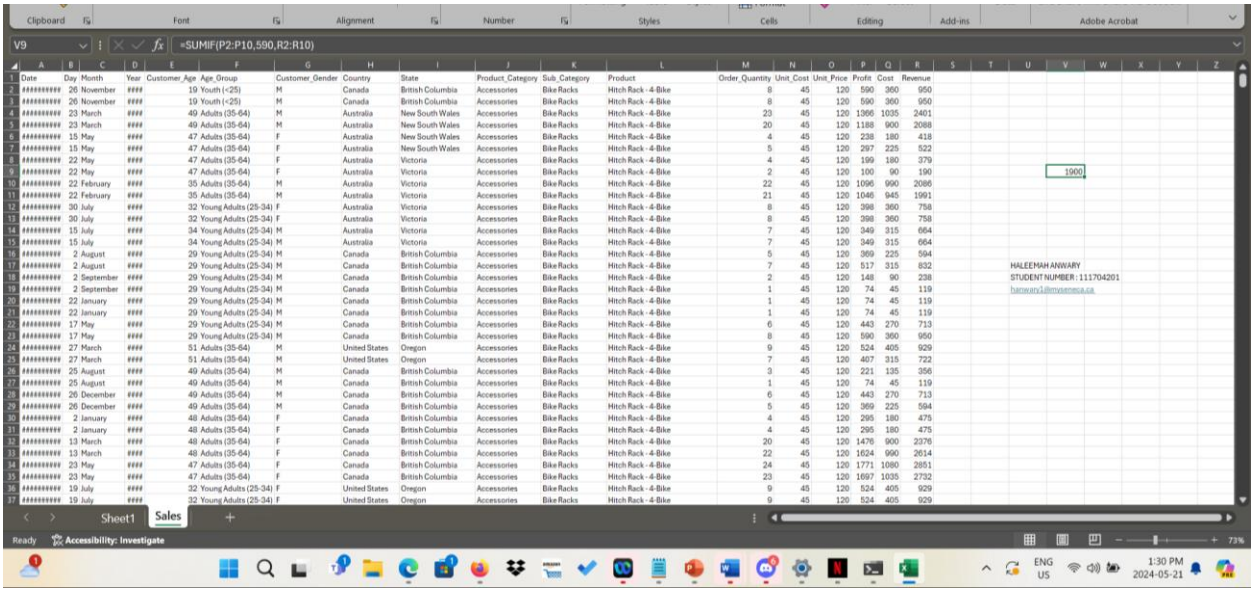
Example 1: In this example I used the formula =SUMIF(P2:P10,">1000",R2:R10) which says find the Sum of the Revenue for Profits over 1000.



Example 2: In this example I used the formula =SUMIF(P2:P10,">1000") which says give sum of profits over 1000.



Example 3: In this example I used the formula =SUMIF(P2:P10,590,R2:R10) which says find the sum of revenue for profits which equal to 590.

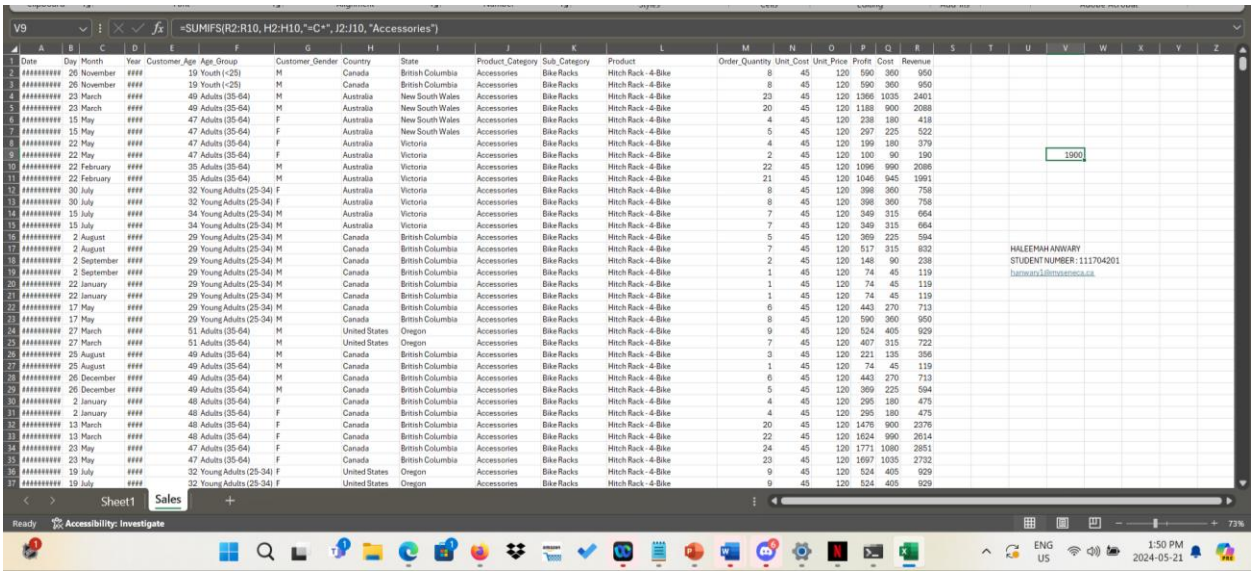


SUMIFS

The SUMIFS function adds all the numbers in a range of cells based on multiple conditions.

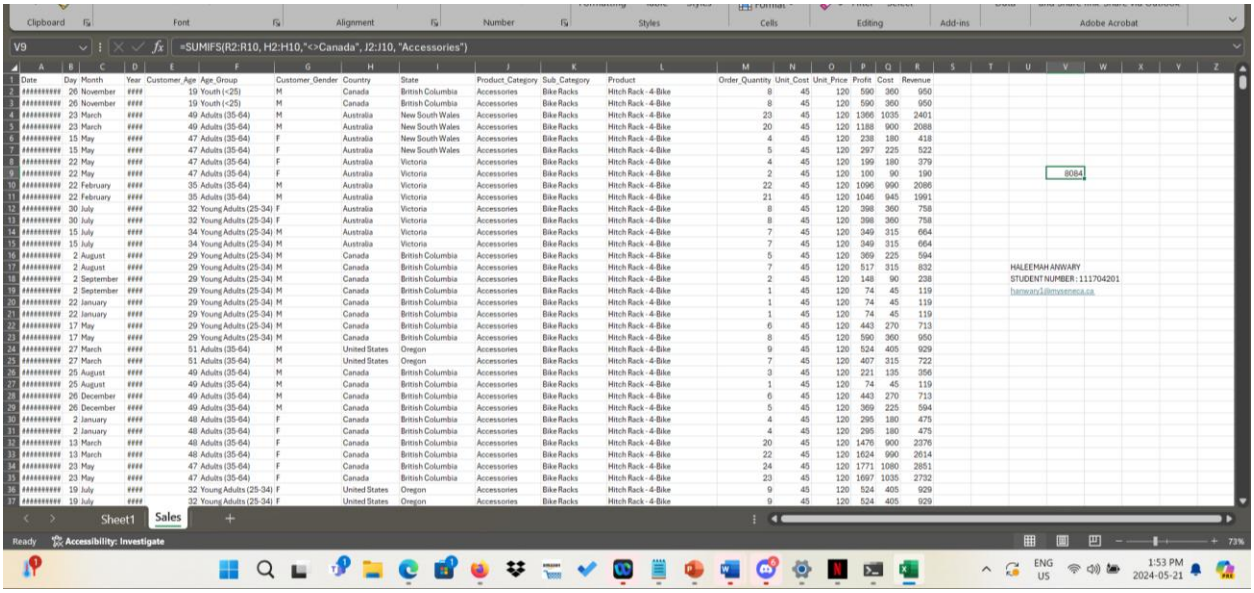
Example 1: In this example I used the formula =SUMIFS(R2:R10, H2:H10,"=C*", J2:J10, "Accessories")

Adds the revenue for countries that begin with ‘C’ and belong to the accessories category. Once it meets both conditions it will give us the result.



Example 2: =SUMIFS(R2:R10, H2:H10,"<>Canada", J2:J10, "Accessories")

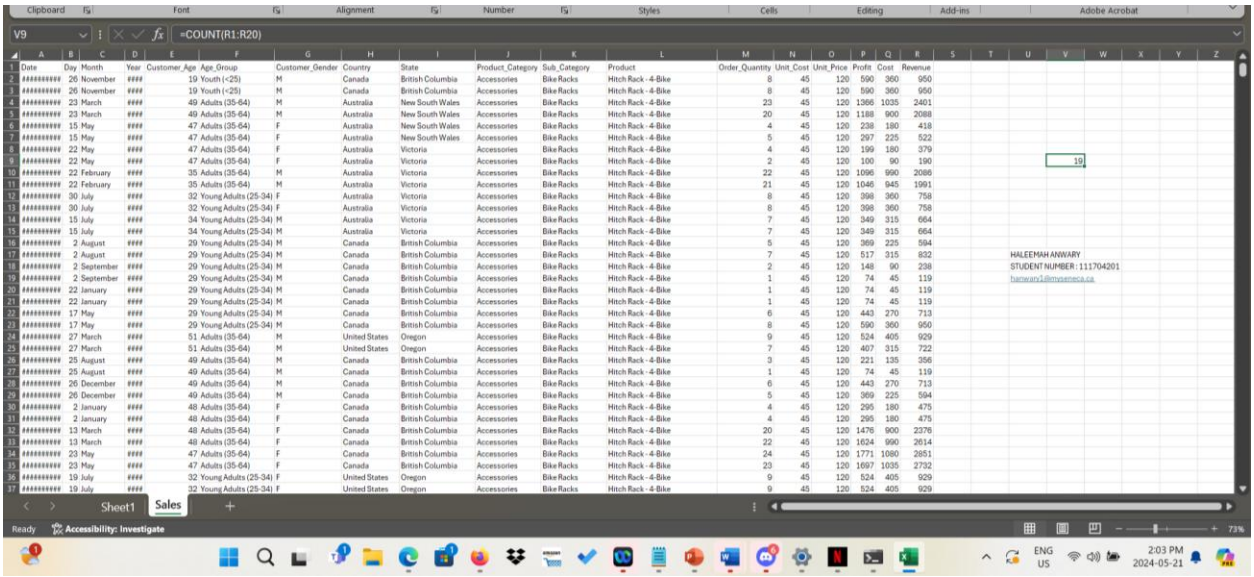
The example is like the above sample, but this excludes “Canada” in the sum.



COUNT

The ‘COUNT’ function counts the number of cells that contain numbers.

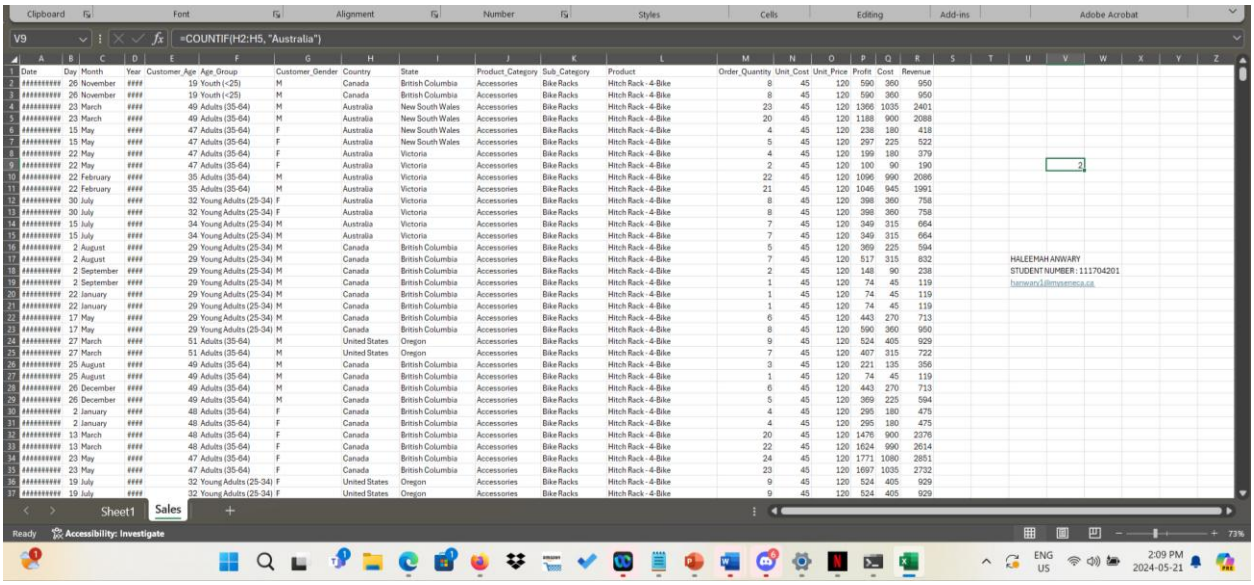
Example 1: I used the formula =COUNT(R1:R20) which counts numeric values In cell range R1:R20.



COUNTIF

The ‘COUNTIF’ function counts the number of cells that meet a single condition.

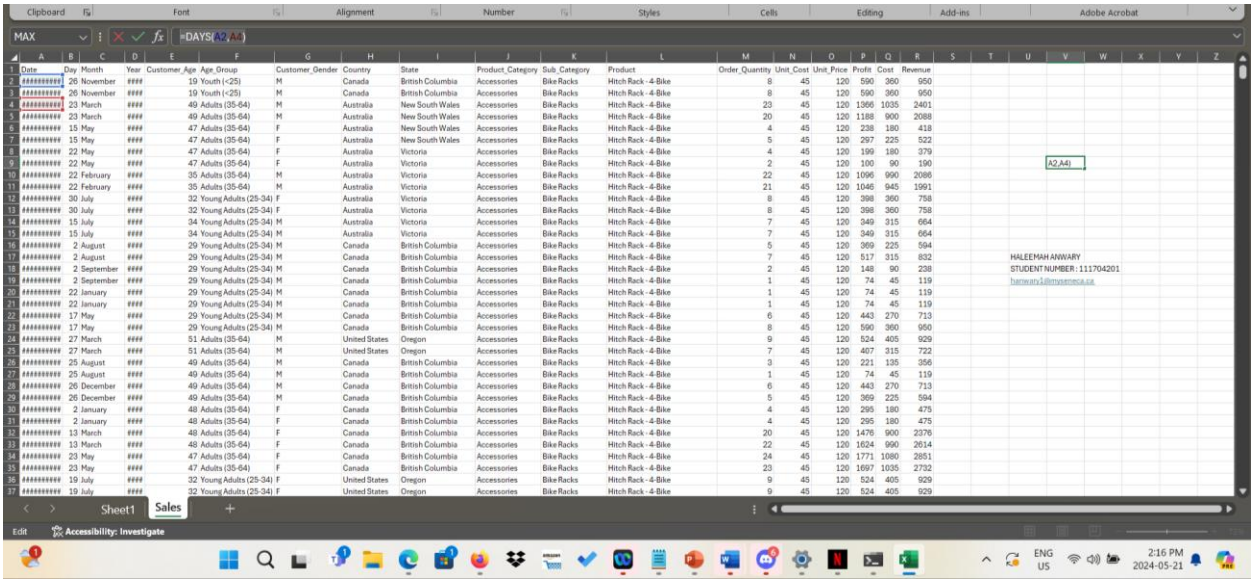
Example 1: In this example I used the formula =COUNTIF(H2:H5, "Australia"). This will give you the count of occurrences of "Australia" in the Country column in the given range.



DAYS

The ‘DAYS’ function returns the number of days between two dates.

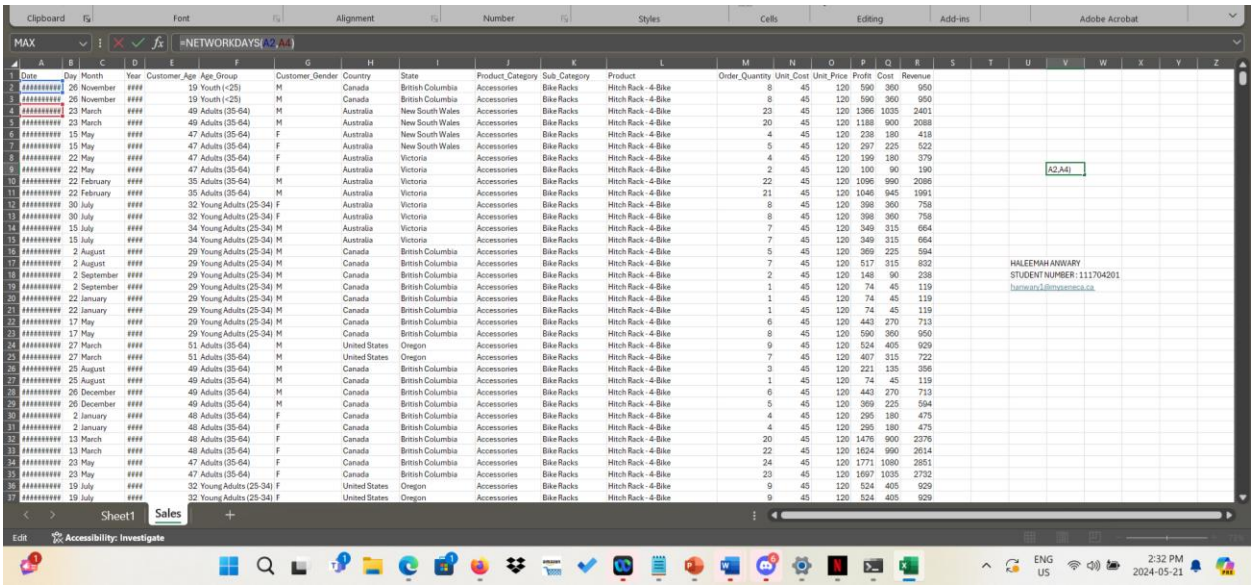
Example: In this formula I used the formula =DAYS(A2,A4) which finds the number of days between the end date in A2 and the start date in A4 .



NETWORKDAYS

The ‘NETWORKDAYS’ function returns the number of whole working days between two dates.

Example: In the example below I used the formula =NETWORKDAYS(A2,A4).

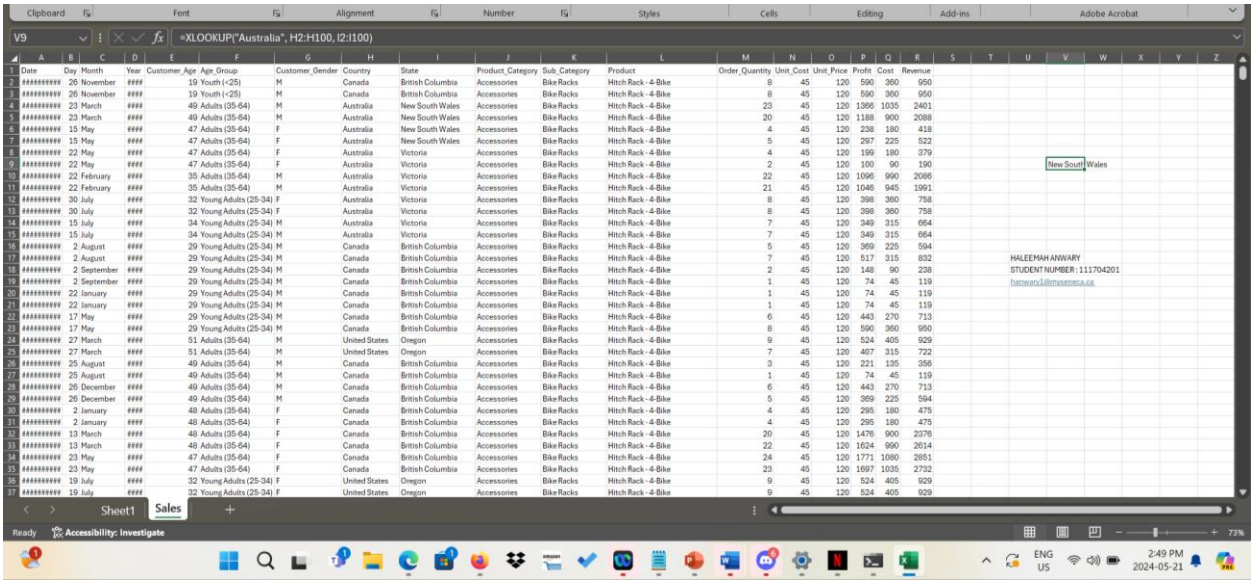


XLOOKUP Tasks (Task 03)

XLOOKUP

The ‘XLOOKUP’ function searches a range or an array and returns an item corresponding to the first match it finds. If a match doesn't exist, then XLOOKUP can return the closest (approximate) match.

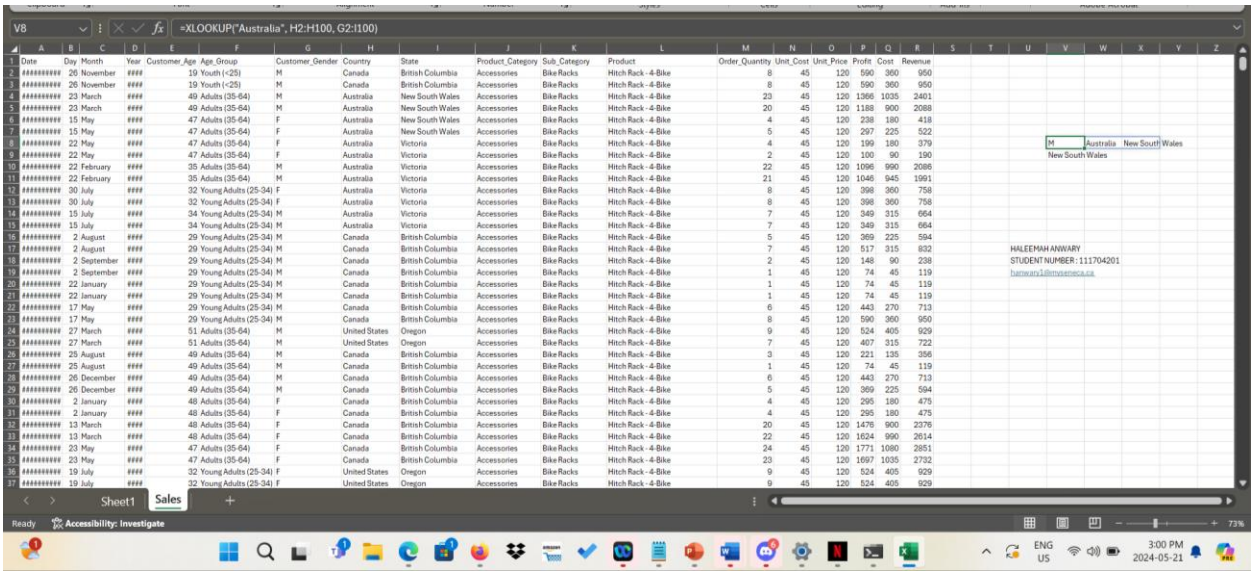
Example: In this example I used the formula =XLOOKUP("Australia", H2:H100, I2:I100). It is used to look up the state based on the customer’s country:



XLOOKUP Multiple Rows

XLOOKUP can be used to return multiple rows of data.

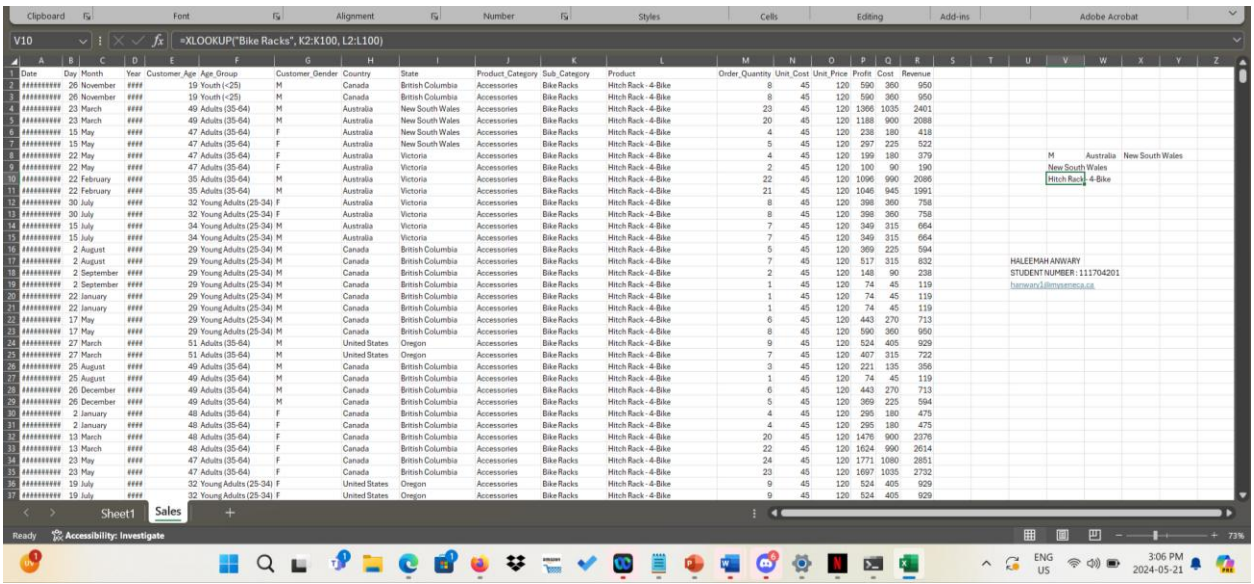
Example: In this example, I am using the formula =XLOOKUP("Australia", H2:H100, G2:I100). formula to look up the state and customer gender based on the customer's country.



XLOOKUP Exact Match

The XLOOKUP function can find an exact match.

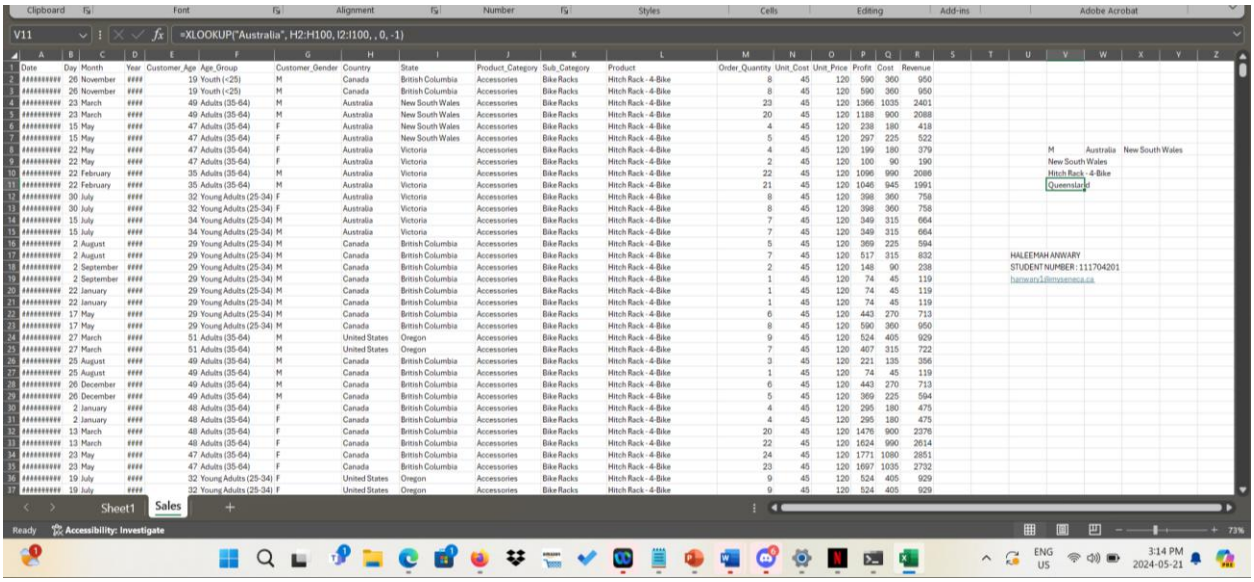
Example: In this example I used the formula =XLOOKUP("Bike Racks", K2:K100, L2:L100) to find the exact match for the product based on the sub-category.



XLOOKUP Search Order

The XLOOKUP function can search in a specific order (first to last or last to first)

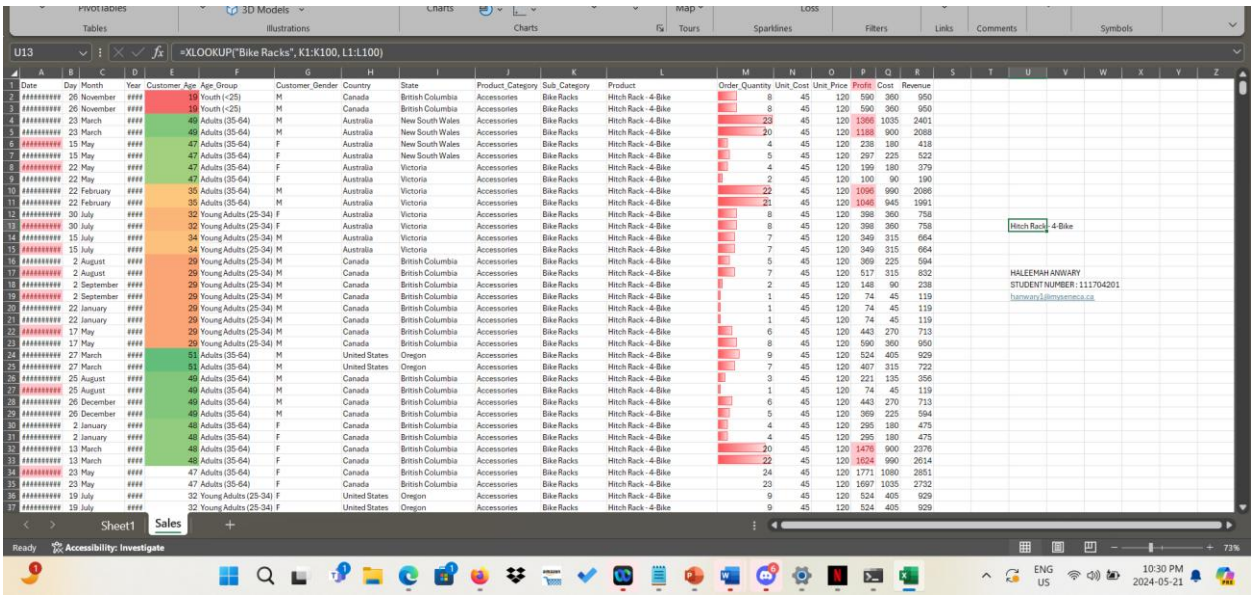
Example: In this example I used the formula =XLOOKUP("Australia", H2:H100, I2:I100, , 0, -1) to find the state searching from the last to the first.



XLOOKUP Horizontal

The XLOOKUP function can search horizontally.

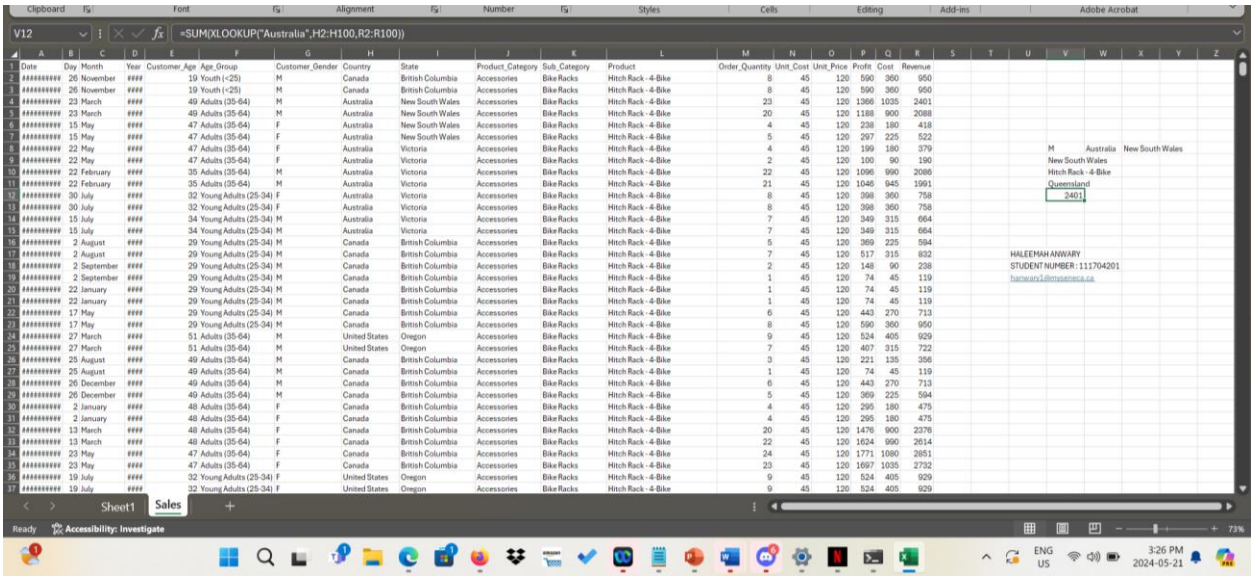
Example 1: In this example I used the formula =XLOOKUP("Bike Racks", K1:K100, L1:L100) to find the product based on the sub-category "Bike Racks" in a horizontal range.



XLOOKUP With Sum

Combine XLOOKUP with SUM-to-sum values based on a lookup.

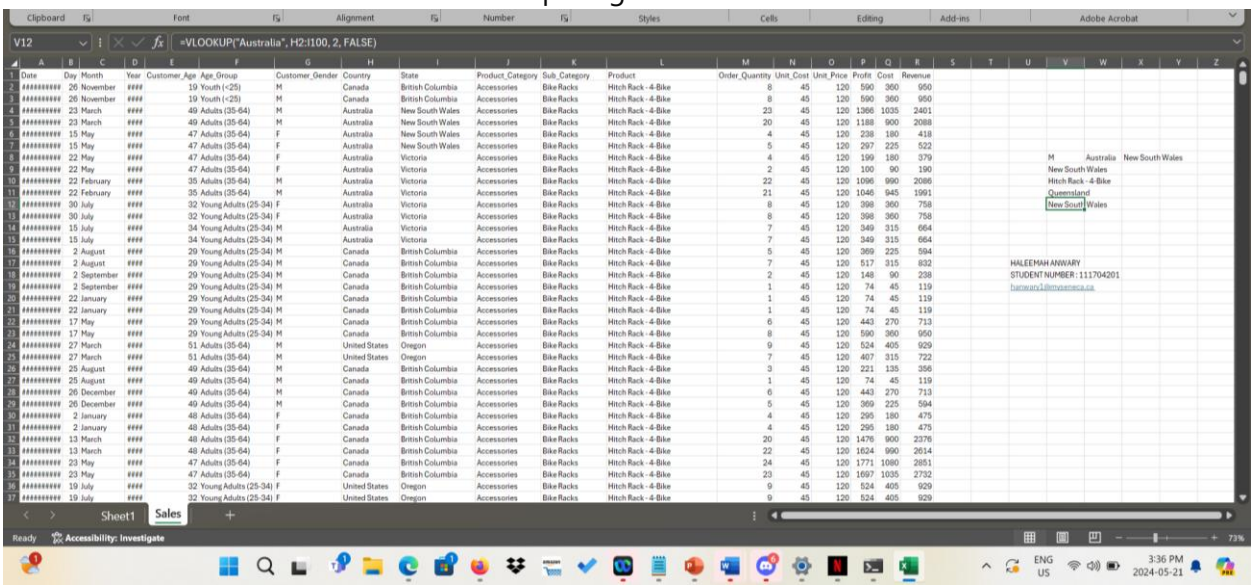
Example: In this example I used the formula =SUM(XLOOKUP("Australia",H2:H100,R2:R100)) to sum the revenue for a specific country.



VLOOKUP

The VLOOKUP function searches for a value in the first column of a range of cells and returns a value in the same row from a specified column.

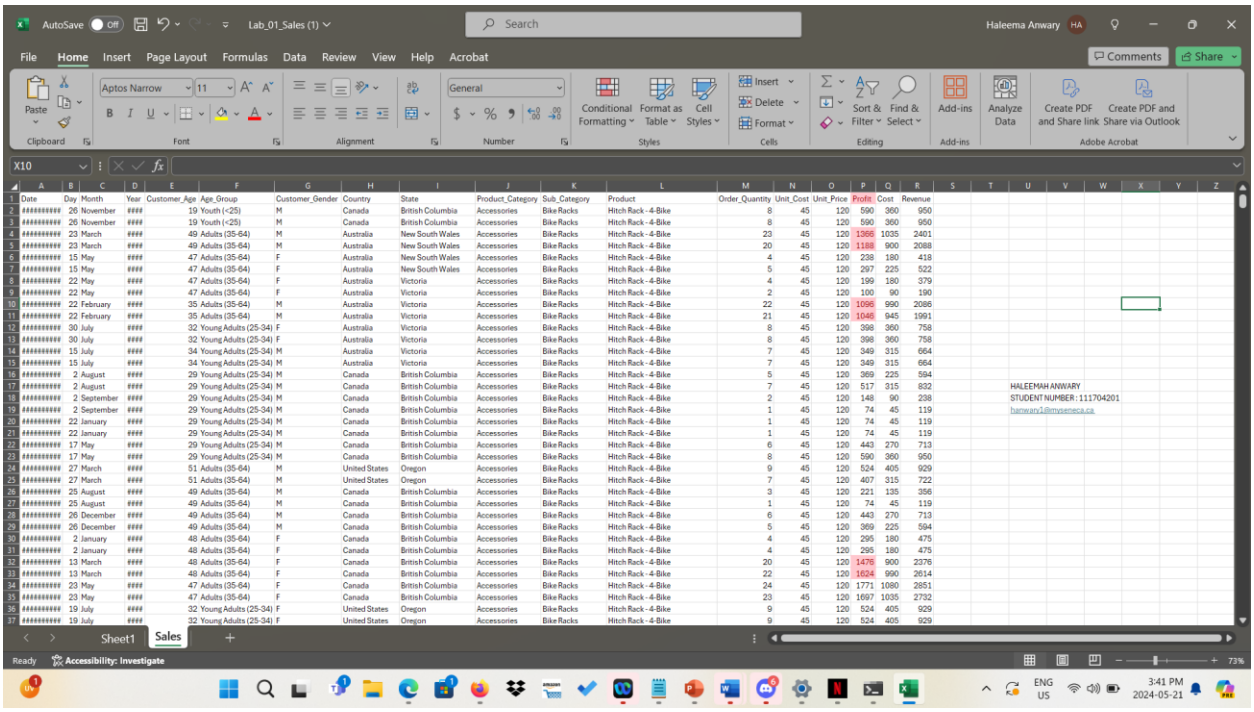
Example: In this example, I used the formula =VLOOKUP("Australia", H2:I100, 2, FALSE) This will return "New South Wales", which is the value in the 2nd column (State) corresponding to the first occurrence of "Australia" in the lookup range.



Conditional Formatting (Task 04)

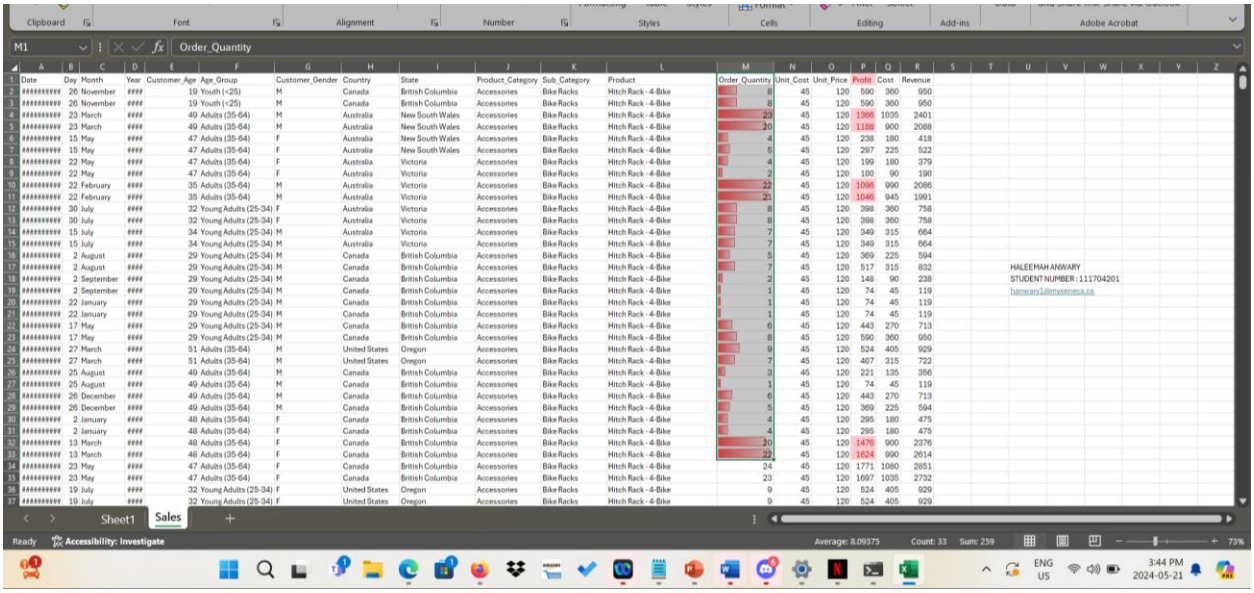
Example 1: Greater than

In this example I've used the Conditional Formatting > Highlight Cell Rules > Greater Than and input 1000 to highlight all cells within the profit column that's greater than 1000 within the specified range.



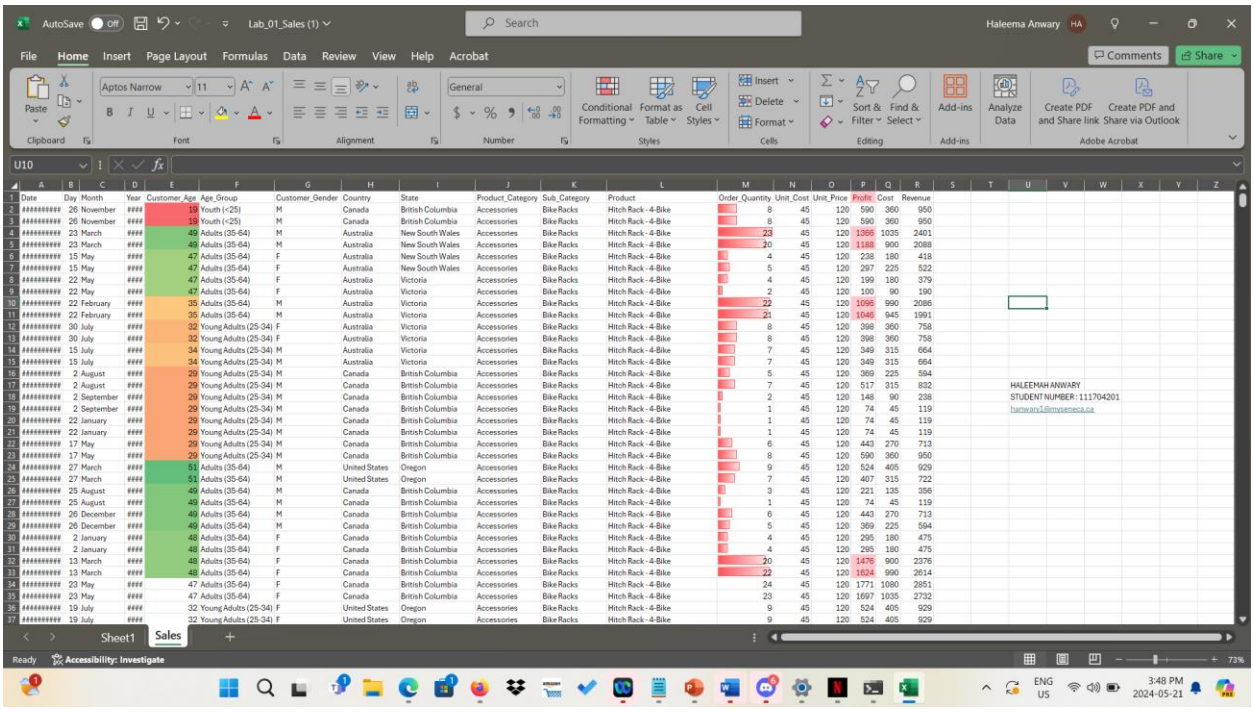
Example 2: Data bars

In this example I've used data bars to represent the order quantity column to make it easier to understand without looking closely at.

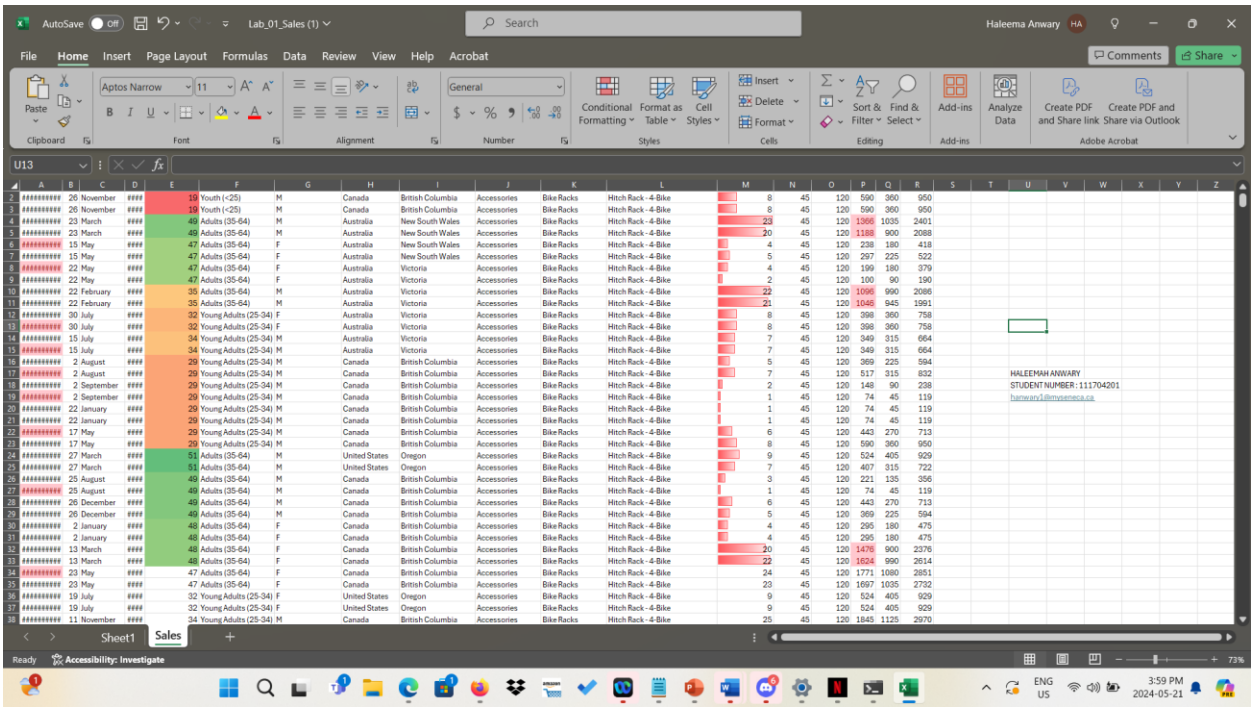


Example 3: Color scales

In this example I've done color scales to represent the customer age column.



Example 4: Between using Date column



References:

Microsoft. (n.d.). *Create a PivotTable to analyze worksheet data*. Retrieved May 21, 2024, from https://support.microsoft.com/en-us/office/create-a-pivottable-to-analyze-worksheet-data-a9a84538-bfe9-40a9-a8e9-f99134456576?ns=excel&version=90&ui=en-us&rs=en-us&ad=us

Microsoft. (n.d.). *Excel functions (alphabetical)*. Retrieved May 21, 2024, from https://support.microsoft.com/en-us/office/excel-functions-alphabetical-b3944572-255d-4efb-bb96-c6d90033e188

Microsoft. (n.d.). *XLOOKUP function*. Retrieved May 21, 2024, from https://support.microsoft.com/en-us/office/xlookup-function-b7fd680e-6d10-43e6-84f9-88eae8bf5929

Microsoft. (n.d.). *Use conditional formatting to highlight information in Excel*. Retrieved May 21, 2024, from https://support.microsoft.com/en-au/office/use-conditional-formatting-to-highlight-information-in-excel-fed60dfa-1d3f-4e13-9ecb-f1951ff89d7f#:~:text=On%20the%20Home%20tab%2C%20in%20the%20Style%20group%2C%20click%20the,and%20then%20select%20a%20format.